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Group Coursework

Group Number - 38

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Contents

1. Experimental Operandum 5
1.1 Group information 5
2. Introduction and Objectives 5
2.1 Independent-sample-test 6
2.2 Paired-sample-test 10
2.3 ANOVA 14
2.4 Pearson Correlation 18
3. Study planning and organization 20
3.1 Experimental Design 21
3.1.1 Experimental Design for Research Question 1 21
3.1.3 Experimental Design for Research Question 3 22
3.1.4 Experimental Design for Research Question 4 (Linushan Sivakumar) 23
3.1.5 Summary Table 25
4. Procedure and Planning 26
4.1 Independent-Sample Test Task - In-Person Moderated 26
4.2 Paired-Sample Test – In-Person (Moderated Unmoderated) Task 27
4.3 ANOVA – Moderated Hybrid Testing 28
4.4 Pearson Correlation – Moderated Remote Testing 31
4.5 Participants Information Sheet 33
5. Roles & Responsibilities 34
5.1 Independent-Sample Test – Roles & Responsibilities 34
5.2 Paired-Sample Test – Roles & Responsibilities 35
5.3 **ANOVA – Roles & Responsibilities** 36
5.4 **Pearson Correlation –Roles & Responsibilities** 39
6. Participant Profile and Recruitment 40
6.1 Independent-Sample Test – Participant Profile and Recruitment 40
6.2 Paired-Sample Test –Participant Profile and Recruitment 41
6.3 ANOVA – Participant Profile and Recruitment 42
6.4 Pearson Correlation –Participant Profile and Recruitment 43
7. Data Collection Methods 44
7.1 **Independent-Sample Test – Data Collection Methods** 44
7.2 Paired-Sample Test –Data Collection Methods 45
7.3. ANOVA – Data Collection Methods 47
7.4 Pearson Correlation – Data Collection Methods 48
8. Research Instruments (materials) and Equipment 50
9. Pilot Testing 59
10. Report the study raw data 60
Ethics form 80
Conclusions 84

Appendix 85
Appendix I. Participant Information Sheet 85
Appendix II. Moderator Information Sheet 88
Appendix III. Participant Recruitment Evidence 88
Appendix IV. Data Collection Material 89
Appendix IV.B **Post-Test Questionnaire Form** 101
Appendix V. Test Recordings 111
115
Appendix V.B. Usability testing session Videos Link 115
Appendix VI. Raw data 116
Appendix VII. Ethics form 116

List of Tables

Table 1- Group Information 5
Table 2 - Research Questions and Hypotheses 6
Table 3 - Independent-sample-test 9
Table 4 - Paired-sample-test 13
Table 5 - Anova test 17
Table 6 - Pearson co-relation test 20
Table 7 - Research Question Summary Table 25
Table 8 - Pre Test Questions Summary Table 52
Table 9 - Post Test Questions Summary Table 54
Table 10 - Post Test Questions Summary Table 55
Table 11 - Post Test Questions Summary Table 57
Table 12 - Post Test Questions Summary Table 59
Table 13 - Pre-Test Questions Raw Data 63
Table 14 - Post Test Questions Raw Data 65
Table 15 - Post Test Questions Raw Data 68
Table 16 - Post Test Questions Raw Data 72
Table 17 - Post Test Questions Raw Data 77

List of Figures

Figure 1- Ethics Form 80

Figure 2 - Ethics Form 81
Figure 3 - Ethics Form 82
Figure 4 - Ethics Form 83
Figure 5 - Participant Information Sheet 86
Figure 6 - Participant Information Sheet 87
Figure 7 - Participant Recruitment Evidence 88
Figure 8 - Participant Recruitment Evidence 89
Figure 9 - Pre-Test Questionnaire Form 90
Figure 10 - Pre-Test Questionnaire Form 91
Figure 11 - Pre-Test Questionnaire Form 92
Figure 12 - Pre-Test Questionnaire Form 93
Figure 13 - Pre-Test Questionnaire Form 94
Figure 14 - Pre-Test Questionnaire Form 95
Figure 15 - Pre-Test Questionnaire Form 96
Figure 16 - Consent Form 97
Figure 17 - Consent Form 98
Figure 18 - Consent Form 99
Figure 19 - Consent Form 100
Figure 20 - Consent Form 100
Figure 21 – Post Study Survey Form 101
Figure 22 – Post Study Survey Form 102
Figure 23 – Post Study Survey Form 102
Figure 24 – Post Study Survey Form 103
Figure 25 – Post Study Survey Form 104
Figure 26 – Post Study Survey Form 105
Figure 27 – Post Study Survey Form 106
Figure 28 – Post Study Survey Form 107
Figure 29 – Post Study Survey Form 108
Figure 30 – Post Study Survey Form 109
Figure 31 – Post Study Survey Form 109
Figure 32 – Post Study Survey Form 110
Figure 33 – Post Study Survey Form 110
Figure 34 – Post Study Survey Form 111
Figure 35 – Post Study Survey Form 112
Figure 36 – Post Study Survey Form 113
Figure 37 – Post Study Survey Form 114
Figure 38 – Post Study Survey Form 115
Figure 39 – Pre Test Form Results 116

1. Experimental Operandum

1.1 Group information

Group Number - 38

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01	**Thalal Izzath**	20200165	w1868568
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03	**Linushan Sivakumar**	20232762	w2084734
04	**Saad Mazhar**	20221804	w2053223

Table 1- Group Information

2. Introduction and Objectives

This experiment will be conducted to assess the usability of the Google Flights platform. The evaluation will focus on its core features for searching, filtering, and selecting travel itineraries, measured in terms of key usability metrics: Effectiveness, Efficiency, Error Tolerance, and Ease of Learning.

As a group, our aim is to identify potential usability problems users encounter during the flight planning process and to understand how these issues impact their overall user experience.

As such, the research objectives are to:

1. Design and conduct a structured usability test to capture both quantitative and qualitative data.
2. Analyze users' interactions with the platform's key functionalities, such as its data-rich search results and advanced filtering tools.
3. Formulate evidence-based recommendations for improving the product's usability.

The present study aims to shed light on the accessibility and usability of Google Flights, particularly for different user groups with varying levels of travel-booking experience.

Research Questions and Hypotheses

Name	Usability Metrics	Statistical Hypotheses Test
Thalal Izzath	Efficiency	Paired-sample-test
Anika Mohamed	Effectiveness	Independent-sample-test
Saad Mazhar	Error Rate	ANOVA
Linushan Sivakumar	Learnability	Pearson Correlation

Table 2 - Research Questions and Hypotheses

Ho = null hypothesis

Ha = alternative hypothesis

μ1 = mean of population 1

μ2 = mean of population 2

μ3=mean of population 3

ρ = Pearson correlation of the population

2.1 Independent-sample-test

Student Name: Anika Mohamed

Research question	Is there a significant difference in the Task Completion Success Rate for finding a mutli-city itinerary between users with prior knowledge of Google Flights and users with no prior knowledge?

Usability metric studied	Effectiveness (measured by task Completion Success Rate)	
Hypothesis (statement)	**Null Hypothesis (H_0):** There is no significant difference in the mean Task Completion Success Rate (μ) for the multi-city search between users with and without prior knowledge of Google Flights.	
	Alternative Hypothesis (H_1): The mean Task Completion Success Rate (μ_1) for users with prior knowledge will be significantly higher than the mean Task Completion Success Rate (μ_2) for users with no prior knowledge.	
Mathematical Expression	**Null Hypothesis (H_0):** $\mu_1 = \mu_2$ μ1 = Mean Task Completion Success Rate of users with prior knowledge. μ2 = Mean Task Completion Success Rate of users without prior knowledge.	
	Alternative Hypothesis (H_1): $\mu_1 > \mu_2$ μ1 = Mean time spent by students with prior knowledge μ2 = Mean time spent by students without prior knowledge	
Variables	Independent	Prior Knowledge (a categorical variable with two levels: 'Yes' or 'No')
	Dependent	Task Completion Success Rate (a binary variable:1 =Success, 0 =Failure)
	Control	All participants will use the Desktop version of Google Flights, be given a common task script, and have a stable internet connection
	Measurement Scale	IV: Nominal DV: Binary (Success/Failure)

	Data type	Categorical/Discrete. The IV is a category (Yes/No) and the DV is a category (Success/Failure)
Experimental design	**Between-Subjects Design (Independent Measure):** The participant's 'Prior knowledge' is a pre-existing attribute, not something the researcher can assign or change. Participants are assigned to a group based on their answer to the pre-test question. This design is the only way to compare two separate, independent groups (those who know the site vs. those who don't to see how first-time use impacts effectiveness.)	
Tasks and Scenarios	**Scenario:** "Imagine you need to plan a multi-city trip. You need to fly from Colombo to London, stay for five days, and then fly from London to New York." **Task:** "Please use Google Flights to find the complete itinerary and the total price for a flight from Colombo to London, and then a second flight from London to New York, five days later."	
Procedure	In-Person Moderated (Lab/Classroom). The moderator must observe the participant and verify the task outcome in real-time to confirm the correct itinerary was found, ensuring a valid measure of the Dependent Variable.	
Data collection method	***Tool 1: Pre-test survey (Google forms)***. This is used to capture the Independent Variable. It is justified as the best fit because it is quick, efficient, and provides clear, documented answers to collect data. ***Tool 2: Post-Test Survey (Google Forms)*** used to capture the Dependent Variable (Success: 1/Failure: 0). The moderator confirms the outcome, which the participant then logs.	
Test material	**Data Collection Instrument:** 1. The pre-test survey contains the key questions: "Have you ever used the Google Flights website before to search for flights? (Yes/No)."	

	2. Post-Test Survey Question includes the success/failure question, time logs (start/end time) **Measurement Protocol:** 1. *Task Start:* Defined as the moment the moderator finishes reading the task and says 'Please begin.' 2. *Task End:* Moment the participant verbally confirms they have found the itinerary and price, and immediately begins filling out the post-test survey. 3. *Measurement of Success (DV):* The moderator confirms the result, and the participant reports the moderator's confirmed outcome (Success/Failure) in the post-test survey.

Table 3 - Independent-sample-test

2.2 Paired-sample-test

Student Name: Thalal Izzath

Research question	**How does the average time taken by users to select a flight itinerary that meets complex criteria** (e.g., maximum one-stop, specific time of the day, cheapest option) **differ when utilizing Google Flights' advanced filtering panel compared to relying solely on analyzing the linear search results list?**	
Usability metric studied	**Efficiency** (to measure efficiency by comparing task completion times under two different conditions)	
Hypothesis (statement)	**Null Hypothesis (H_0):** The study aims to determine if there is a significant difference in the average time required for users to successfully identify and select a flight itinerary that meets complex, predefined criteria when they use the advanced filtering panel on Google Flights versus when they rely on manual scanning and basic filtering of the linear search results.	
	Alternative Hypothesis (H_1): The average amount of time required to select a complex flight itinerary differs significantly when users utilize the advanced filtering panel compared to when they rely on the linear search results list alone.	
Mathematical Expression	**Null Hypothesis (H_0): $\mu_1 = \mu_2$** μ_1 = Mean time spent by students with prior knowledge μ_2 = Mean time spent by students without prior knowledge	
	Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ μ_1 = Mean time spent by students with prior knowledge μ_2 = Mean time spent by students without prior knowledge	
Variables	Independent	Method of Itinerary Selection. This involves two conditions: - Condition 1: Relying on the linear search results list and basic sorting.

		- Condition 2: Using the advanced filtering panel to find the complex itinerary.
	Dependent	Time taken (in seconds/minutes) for the user to successfully select a flight itinerary that meets all complex criteria.
	Control	Participants focus on completing the same complex itinerary selection task (e.g., same city pair, travel dates, and exact flight criteria)
	Measurement Type	Ratio Scale: Used for Task Completion Time. Time measured in seconds or minutes has a true zero point and equal intervals Secondary - Ordinal Scale: Used for the Perceived Efficiency/Ease ratings collected via the post-test survey questions (e.g., Likert scale from 1 to 5, where 5 is "Very Efficient")
	Data type	**Quantitative Data (Continuous):** The time taken is measured numerically (in seconds/minutes), making it a continuous quantitative variable
Experimental design	**Within-Subjects Design** The fundamental reason for choosing a Within-Subjects Design (Paired-Sample) is to minimize the influence of individual differences among participants on the outcome measures In a Within-Subjects Design, every participant performs tasks under all conditions being tested. In my case, this means every participant will attempt the complex itinerary selection task on Google Flights using both: **1. Condition 1 (Linear Search Results List/Basic Filtering) (μ).** **2. Condition 2. (Advanced Filtering Panel) (μ1).**	

	By using this approach, I wanted to effectively compare each participant's performance (time taken) to their own baseline performance
Tasks and Scenarios	The participant must locate and select a specific flight itinerary using Google Flights based on a challenging scenario (e.g., specific dates, a round trip between London and New York, restricted by budget and flight features). The complexity of the criteria is essential to ensure that users must engage with the filtering interface. **Condition A: Relying on the Linear Search Results List (μ2)** Objective: To measure the time taken to achieve the same success when the user is forced to manually scan, scroll, and rely on basic sorting rather than advanced filters. *"Without using the advanced filtering panel (e.g., avoid using the detailed layover duration, specific airline alliance, or multi-slider range tools), find and select the cheapest return flight from [City X] to [City Y] within a 7-day window that includes no more than one stop. You must rely primarily on scanning the linear search results.* **Condition B: Using the Advanced Filtering Panel (μ1)** Objective: To measure the time taken to achieve success when the user utilizes the full range of efficiency-enhancing filtering tools. *"Use the advanced filtering panel options (such as stop duration slider,, flight duration slider, and aircraft type filter) to find and select the cheapest return flight from [City A] to [City B] within a 7-day window that includes no more than one stop."*
Procedure	The Paired-Sample Test requires maximum control to ensure the two conditions are the only factors causing a difference in the outcome. Therefore, **the In-Person Moderated Task is the ideal** choice for measuring efficiency, as it allows to: 1. Use a controlled lab environment for accurate time measurement (ratio scale data).

	2. Standardize the task delivery and guarantee participants comply with the specific constraints of the filtering tasks (μ1 vs. μ2). 3. Collect both objective quantitative data (time) and subjective quantitative data (perceived efficiency via post-test surveys)
Data collection method	The digital timer is the key instrument used to collect objective, quantitative data for the Efficiency metric. The Paired-Sample Test requires precise measurement of the time taken for each participant to complete the complex itinerary search task under two different conditions.
Test material	**Materials -** **1. Task Instructions for Participants** *Structured instructions provided in document format (Appendix II) to guide participants through both comparative tasks (Advanced Filtering vs. Basic List Search)* **2. Questionnaire to gather demographic information and prior knowledge** *The Pre-Test Questionnaire Form (Appendix IV.A) used to collect basic demographic data, academic background, and assess prior experience/familiarity with the application.* **Measurement -** **1. Task Start** *The task starts when the participant begins interaction with the interface.* **2. Task End** *The task ends upon successful task selection of the flight itinerary that meets all required criteria.*

Table 4 - Paired-sample-test

2.3 ANOVA

Student Name: Saad Mazhar

Research question	Do users with different levels of flight-booking experience (novice, intermediate, and expert) differ significantly in the number of errors they make when completing a flight-search task on Google Flights?	
Usability metric studied	**Error Tolerance** **(**Measured through; Error Count, the total number of mistakes a participant makes while attempting to find a flight that matches all required criteria.)	
Hypothesis (statement)	**Null Hypothesis (H_0):** There is no significant difference in mean error rate among novice, intermediate, and expert users on Google Flights.	
	Alternative Hypothesis (H_1): There is a significant difference in mean error rate between at least two of the experience groups.	
Mathematical Expression	**Null Hypothesis (H_0):** $\mu_1 = \mu_2 = \mu_3$	
	Alternative Hypothesis (H_1): At least one group mean differs: $\mu_1 \neq \mu_2$ or $\mu_1 \neq \mu_3$ or $\mu_2 \neq \mu_3$ where: μ_1 = mean number of filters used by novice participants μ_2 = mean number of filters used by intermediate participants μ_3 = mean number of filters used by expert participants	
Variables	Independent	User experience level (three categories): ● Novice (0–2 previous bookings) ● Intermediate (3–10 previous bookings) ● Expert (>10 previous bookings)
	Dependent (DV)	Error Count (total number of errors recorded using a structured error checklist.)
	Control	Same travel route (Sri Lanka → London)

		Same specific dates (10–17 December) Same budget limit (£250) Same stopover restriction (maximum one stop) Same device, browser, and lab environment Standardised task instructions
	Measurement Scale	Error Count: Ratio scale (0, 1, 2, 3 …)
	Data type	Quantitative (Continuous) number of errors recorded from the error checklist.
Experimental design	**Between-Subjects Design (One-Way ANOVA)** Each participant belongs to only one experience group. This design eliminates learning effects and ensures that differences in error rate reflect prior experience rather than task repetition.	
Tasks and Scenarios	**Scenario:** *You need to book a round-trip flight from Sri Lanka to London between 10–17 December, staying within a £250 budget, and choosing a flight with no more than one stop.* **Task:** *Participants must use Google Flights to find a flight itinerary that matches all the given criteria. Error Count is recorded while they navigate the interface.* The **same task** is given to all participants to ensure the comparison across groups is valid for ANOVA.	
Procedure	The data collection was carried out using a moderated hybrid approach, involving both in-person and remote sessions: **In-Person Moderated Sessions**	

	Conducted with participants at: ● Informatics Institute of Technology (IIT) ● Information and Communication Technology Agency (ICTA) of Sri Lanka ● Spectrify AI **Remote Moderated Sessions** Conducted via Zoom/Google Meet with participants at: ● IFS ● LSEG ● Additional remote participants (colleagues/friends) In both modes, participants shared their screens while performing the task. The moderator observed without giving help, recorded errors in real time, and verified them later using OBS recordings. **Procedure Steps:** 1. Introduction (5 min) - Briefing + consent collection. 2. Pre-task survey (3 min) - Determines experience level (Novice/Intermediate/Expert). 3. Main Task (10–12 min) - Participants complete the flight-search task independently. ● Moderator logs errors using Error Checklist ● OBS Studio used to verify errors afterwards ● No guidance or hints provided 4. Post-task questionnaire (5 min) - Participants rate difficulty, clarity, and provide brief feedback. 5. Debrief (2–3 min) - Participants are thanked and reminded of confidentiality. Total session duration: ~15–30 minutes.

Data collection method	**Tools Used:** ● **OBS Studio (Screen Recording)** - Captures all actions to verify errors. ● **Error Checklist** - Used by the moderator to log every mistake made. ● **Digital Timer** - Records task start and end times. ● **Google Forms** - Used for the pre-task survey and post-task questionnaire. ● **Observation Sheet** - used to record errors in real-time and later verified with recordings.
Test material	Materials: ● Structured task instruction sheet (Appendix II) ● Pre-task survey (Appendix III) ● Error checklist used during observation (Appendix IV) ● Post-task questionnaire (Appendix V) Measurement Definitions: ● Task Start: When the participant clicks "Search" after entering the flight details ● Task End: When the participant verbally confirms they have selected a valid itinerary ● Error Count: Number of mistakes recorded from the predefined error checklist This Error Count is the dependent variable tested using ANOVA.

Table 5 - Anova test

2.4 Pearson Correlation

Student Name: Linushan Sivakumar

Research question	Is there a relationship between the reduction in task completion time from a first attempt to a second attempt at applying advanced filters on Google Flights and the user's post-test confidence rating in using those filters?	
Usability metric studied	Ease of Learning	
Hypothesis (statement)	**Null Hypothesis (H_0):** There is no significant correlation ($\rho=0$) between the reduction in task completion time from a first attempt to a second attempt at applying advanced filters on Google Flights and the user's post-test confidence rating in using those filters.	
	Alternative Hypothesis (H_1): There is a significant correlation ($\rho \neq 0$) between the reduction in task completion time and the user's post-test confidence rating in using advanced filters on Google Flights.	
Mathematical Expression	**Null Hypothesis (H_0): $\rho = 0$** Where ρ = Pearson Correlation coefficient	
	Alternative Hypothesis (H_1): $\rho \neq 0$ Where ρ = Pearson Correlation coefficient	
Variables	**Independent**	Time Reduction Score. This score is a single measure representing the magnitude of learning.
	Dependent	The user's post task confidence rating in using advanced filters.
	Control	The specific task of applying the same set of advanced filters on Google Flights

	Measurement Scale	Time Reduction Score is measured on a Ratio scale and Confidence Rating is measured on an Ordinal scale, treated as continuous.
	Data type	Time Reduction Score : Continuous Quantitative Data Confidence Rating : Ordinal Scale
Experimental design	**Within-Subjects Design** ● Ensures that every participant is exposed to both key conditions necessary to measure Learnability: the first attempt at the task and the second attempt at the task ● All participants are exposed to the same usability tasks for a direct comparison of Learnability metrics.	
Tasks and Scenarios	Participants will 1. Complete the task **unguided** for the first time. 2. Read a step-by-step guide/tutorial for the task. Time is not recorded here. 3. Complete the exact same task again, unguided. 4. Fill out the Post-Test Questionnaire (Confidence Rating).	
Procedure	**Type :** Moderated Remote testing **Justification :** The session is guided to ensure the tutorial is delivered consistently during both attempts. Conducting the test remotely enhances participant convenience and helps recruit a larger, more diverse group of users.	
Data collection method	● A screen recording software like OBS Studio to watch user activity to get an accurate, timestamped, objective record of the task start and end points for both attempts. ● An Online survey tool like Google forms to gather satisfaction scores (Likert scale) for systematic data collection ensuring participant confidentiality and simplifies data export for analysis.	

Test material	**Materials :** 1. A Structured task instruction for the participant 2. A post-task questionnaire to gather participant rating **Measurement :** 1. The task starts when the participant clicks the first advanced filter element after flight search results are loaded. 2. Task end when the screen stabilizes showing the final itinerary and price meeting all criteria 3. Data : Reduction time in seconds for correlation analysis

Table 6 - Pearson co-relation test

3. Study planning and organization

The study was organised as a single, standardised remote-moderated session designed to collect data for all four research questions efficiently. A unified Test Session Flow was developed, requiring each participant to complete the consent form, pre-test survey, assigned tasks, and the post-test survey in one 20–25 minute session.

Each research question followed its own experimental design (between-subjects or within-subjects), yet all procedures were integrated into a coherent sequence to avoid interference between variables. Group members individually moderated their respective tasks but adhered to a shared script, consistent task instructions, and uniform data-collection tools, including timing sheets, error checklists, and screen-recording verification.

As a group, we collaboratively prepared recruitment materials, survey instruments, task scenarios, and moderation protocols to ensure methodological consistency. This coordinated structure enabled reliable, comparable data collection across participants and supported the validity of subsequent statistical analyses.

3.1 Experimental Design

Our study was structured as a moderated test session that integrated all four required statistical designs. Each of the 20 participants completed the session one by one, and the Test Session Flow was built to seamlessly collect the data for each group member's hypothesis.

This integrated approach allowed us to efficiently gather data for different types of comparisons (between-groups and within-groups) in one 30-45 minute session.

3.1.1 Experimental Design for Research Question 1

Student Name: Anika Mohamed

This research question employed a between-subjects design to examine the effect of prior knowledge on the effectiveness of completing a multi-city search task in Google Flights**.** This design was chosen to prevent carryover effects and ensure that each participant's first attempt accurately reflected their natural task performance.
The independent variable (IV) was prior knowledge (participants with vs. without prior experience using Google Flights), and the dependent variable (DV) was task completion success (coded as success = 1, failure = 0). Participants were asked to plan a multi-city trip, booking a flight from Colombo to London, staying five days, and then flying from London to New York using Google Flights to find the full itinerary and total cost.

An In-Person Moderated procedure was used to ensure data fidelity. A group member acted as the moderator to:

1. Administer the pre-test survey, thereby identifying the participant's prior knowledge status (the IV).
2. Verify each participant's task outcome via screen sharing in real-time.

This approach guaranteed reliable data collection and accurate measurement of the Task Completion Success Rate between the two independent user groups, providing the necessary data foundation for comparative statistical analysis.

3.1.2 Experimental Design for Research Question 2

Student Name: Thalal Izzath

A within-subjects design (Paired-Sample) was used to assess how utilizing Google Flights' advanced filtering panel impacts user efficiency during the itinerary selection process.

Each participant performed tasks under two conditions: using the advanced filtering panel and relying on the linear search results list/basic filtering. This paired-sample approach was chosen to minimize individual differences and focus differences in time solely on the effect of the filtering method on efficiency.

The independent variable was the Method of Itinerary Selection (Advanced Filter vs. Basic List). The dependent variable was Efficiency, primarily measured by the time taken to select the itinerary, and secondarily through perceived efficiency ratings (Ordinal Scale). Participants completed two tasks, one using the advanced filtering panel and one using the basic list/sorting, and their time metrics were compared.

The experiment was conducted as an In-Person Moderated Task in a controlled lab environment to minimize distractions. Moderators ensured participants received consistent instructions and strictly adhered to the feature usage constraints. Data was collected using a digital timer for objective time measurement and post-test surveys for perceived efficiency ratings, capturing both quantitative and qualitative insights. This design allowed for a direct comparison of Efficiency between the two filtering methods, aligning with the research question's objectives.

3.1.3 Experimental Design for Research Question 3

Student Name: Saad Mazhar

This study used a between-subjects experimental design structured for a one-way ANOVA to examine whether users with different levels of flight-booking experience (novice, intermediate, expert) differ in the number of errors they make when completing a standardised Google Flights search task. A between-subjects approach was chosen to avoid learning effects and ensure that each participant's performance reflected their natural familiarity with flight-search tools.

The **independent variable (IV)** was experience level, based on the number of past flight bookings reported in the pre-test survey. The **dependent variable (DV)** was Error Count, recorded using a predefined checklist that captured mistakes such as selecting incorrect dates, exceeding the budget, choosing too many stops, or navigating away from relevant pages.

Data was collected through a combination of in-person moderated testing (IIT, ICTA, Spectrify AI) and remote moderated sessions (participants at IFS, LSEG and others). All moderators followed the same script, observed without assisting, and logged errors in real time. Screen recordings were used to verify accuracy.

Every participant completed the same task scenario: *"Find a round-trip flight from Sri Lanka to London between 10–17 December, within a £250 budget, and with no more than one stop."*

Using a fixed scenario ensured that differences in error rates were due to user experience rather than task complexity. Overall, this design provided a controlled and reliable basis for comparing performance across the three experience groups and generating valid quantitative data for one-way ANOVA analysis.

3.1.4 Experimental Design for Research Question 4 (Linushan Sivakumar)

The chosen experimental design for this study is a Within-subjects design. This design is most appropriate for testing the hypothesis related to Learnability because it directly compares each participant's performance against their own baseline.

Within-subjects design allows the measurement of the core variable, Reduction in Task Completion Time(Time Attempt $_1$ $-$ Time Attempt $_2$), by testing the same individual under both conditions. This direct comparison is essential to accurately quantify the learning effect triggered by the intervention. This design also eliminates individual differences as a confounding factor. Since each participant completes both the initial and final task attempts, they serve as their own control. This ensures that observed changes in performance are

attributable to the learning intervention rather than differences in prior experience, technical skill, or other external factors.

Additionally, the within-subjects design facilitates correlation analysis. Both key variables for the Pearson correlation, the continuous time reduction score and the confidence rating are derived from the same participant, ensuring data consistency and reliability for the analysis.

Thus, the study is structured to collect data efficiently and validly by having all participants complete both task attempts with a learning intervention in between, followed by a confidence rating assessment to address the formulated hypothesis effectively.

3.1.5 Summary Table

Research Question	**Experimental Design**	**Independent Variable**	**Dependent Variable**	**Participants' Tasks**
Is there a significant difference in the Task Completion Success Rate for finding a mutli-city itinerary between users with prior knowledge of Google Flights and users with no prior knowledge?	Between-Subjects Design	Prior Knowledge of using google flights	Task Completion Success Rate	use Google Flights to find the complete itinerary and the total price for a flight from Colombo to London, and then a second flight from London to New York, five days later
How does the average time taken by users to select a flight itinerary that meets complex criteria (e.g., maximum one-stop, specific time of the day, cheapest option) differ when utilizing Google Flights' advanced filtering panel compared to relying solely on analyzing the linear search results list?	Within-Subjects Design	The participant must locate and select a specific flight itinerary using Google Flights based on a challenging scenario.	Time taken (in seconds/minutes) for the user to successfully select a flight itinerary that meets all complex criteria	locate and select a specific flight itinerary using Google Flights based on a challenging scenario (e.g., specific dates, a round trip between London and New York, restricted by budget and flight features). The complexity of the criteria is essential to ensure that users must engage with the filtering interface.
Do users with different levels of flight-booking experience (novice, intermediate, and expert) differ significantly in the number of errors they make when completing a flight-search task on Google Flights?	Between-Subjects Design	User experience level (Novice, Intermediate, Expert)	Error Count	Participants use Google Flights to find a flight itinerary that meets all predefined criteria. The moderator records each error during the task using an error checklist.
Is there a relationship between the reduction in task completion time from a first attempt to a second attempt at applying advanced filters on Google Flights and the user's post-test confidence rating in using those filters?	Within-Subjects Design	Time Reduction Score	user's post task confidence rating in using advanced filters	Complete the task unguided for the first time. Read a step-by-step guide/tutorial for the task. Time is not recorded here. Complete the exact same task again, unguided and fill out the Post-Test Questionnaire.

Table 7 - Research Question Summary Table

4. Procedure and Planning

4.1 Independent-Sample Test Task - In-Person Moderated

Student Name: Anika Mohamed

The research question was addressed through an In-Person Moderated approach, conducted within a controlled lab or classroom environment. This procedure was selected to ensure the accurate identification of the participant's group (Prior Knowledge, the IV) and the precise verification of the complex task outcome (Success/Failure, the DV), while minimizing external variables and distractions associated with remote testing.

Procedure:

1. Participants were assigned to one of two independent groups (Prior Knowledge vs. No Knowledge) based on their self-reported response to a pre-test survey question (estimated duration: 4–5 minutes).
2. The session was conducted physically in a lab environment to control distractions. The moderator sits nearby to observe the participant..
3. During the in-test phase (approximately 15–20 minutes), the moderator began by reading the multi-city travel scenario and the task instructions aloud to the participant. The assigned task involved planning a three-leg itinerary using Google Flights with the following routes and dates:
    - Leg 1: CMB → LHR on March 10, 2026
    - Leg 2: LHR → JFK on March 15, 2026
    - Leg 3: JFK → CMB on March 20, 2026
4. The moderator observed the participant's interaction in real-time throughout the task.
5. Upon verbal confirmation of task completion, the moderator visually verified the output against a pre-defined checklist to confirm the itinerary was correct (DV**:** Success/Failure). The participant then immediately completed the post-test survey, logging their start/end times and confirming the verified success/failure outcome.

Justification:

The in-person moderated setup provides the highest level of data integrity for this experiment. By conducting the test in a controlled lab, environmental variables are reduced. Crucially, moderation allows the researcher to:

1. Visually verify the complex, multi-step output in real-time before marking the binary outcome, eliminating uncertainty inherent in unmoderated testing.
2. Ensure adherence to the standardized instructions and task requirements.
3. Accurately confirm the participant's group status (IV) based on the pre-test before starting the task.

Data Collection Tools and Materials:

1. Google Forms (Pre-Test Survey): Used to gather participants' prior knowledge, which serves as the Independent Variable.
2. Google Forms (Post-Test Survey): Used to log the Dependent Variable (Success/Failure).
3. Lab/Classroom Environment: Used as a controlled setting to ensure standardized testing conditions.
4. Standardized Task Instructions: Provided to the moderator to maintain consistency in how the task and scenarios are presented across all sessions.

4.2 Paired-Sample Test – In-Person (Moderated Unmoderated) Task

Student Name: Thalal Izzath

For the research question concerning the difference in efficiency when utilizing Google Flights' advanced filtering panel versus the basic search list, a paired-sample design was implemented using an in-person moderated approach.

Procedure

Participants performed two comparative tasks corresponding to the within-subjects design:

1. **Task 1 (Condition A):** Participants did the task but were constrained to rely on the linear search results list and basic sorting without using the advanced filtering features.
2. **Task 2 (Condition B):** Participants searched for and selected a complex flight itinerary utilizing the advanced filtering panel.

Efficiency metrics such as task completion time (ratio scale data) and post-task survey ratings on perceived efficiency (ordinal data) were recorded for both tasks. Following the precedent of the sample, each task took 5–10 minutes, with an additional 10 minutes for pre-task instructions and post-task feedback, resulting in a total session duration of approximately 25 minutes..

Justification

An in-person moderated design was selected to ensure participants could consistently complete both tasks while mitigating potential errors. Moderation provided the opportunity to standardize instructions and was essential for observing participants in real-time to verify that they adhered to the specific usage constraints of the filtering features in each condition.

Data Collection Tools and Materials

1. Task instructions sheet (Google Document) and post-task surveys (for perceived efficiency ratings).
2. Digital Timer to accurately measure task completion time (the primary efficiency metric).
3. Task setup in the Google Flights environment.

Measurement

• Task Start: The task starts when the participant begins interaction with the interface.
• Task End: The task ends upon successful selection of the flight itinerary that meets all required criteria.

4.3 ANOVA – Moderated Hybrid Testing

Student Name: Saad Mazhar

A Moderated Hybrid Testing approach was employed to investigate whether users with different levels of flight-booking experience (Novice, Intermediate, Expert) differ in their Error Rates when completing a standardised Google Flights task. The combination of in-person and remote moderated sessions enabled broad participant accessibility while ensuring consistent task delivery, accurate error observation, and control over testing conditions across all groups.

Procedure: Between-Subjects Session Flow

All participants completed the same flight-search scenario, and their assigned experience level (IV) was confirmed through the pre-test survey. The study followed a structured four-phase procedure:

1. **Briefing & Setup (Moderator-Led)**
   Participants were welcomed and briefed on the task scenario: *"Find a round-trip flight from Sri Lanka to London (10–17 December), under £250, with no more than one stop."*
   Screen sharing was enabled for remote users or a workstation was provided in person.

2. **Task Execution (Unassisted Performance)**
   Participants completed the task independently.
   The moderator:
   - Observed the interaction in real time
   - Logged mistakes using a predefined Error Checklist
   - Provided no guidance or corrections

   This phase produced the dependent variable: Error Count.

3. **Post-Task Reflection**
   Participants completed a short post-test survey capturing perceived difficulty, clarity of instructions, and issues encountered.

4. **Session Closure**
   The moderator thanked the participant and saved observation notes and screen recordings for later verification.

Session Duration: ~15–25 minutes per participant.

Justification

The hybrid moderated approach ensured both ecological validity and methodological control. Moderation allowed the researcher to:

- Deliver identical instructions across all participants
- Accurately record errors as they occurred
- Validate observations through screen recordings

Despite the presence of a moderator, task completion remained fully unassisted, ensuring that the Error Rate genuinely reflected the participant's natural performance and differences attributable to experience level. This level of control is essential for producing valid, comparable data for the one-way ANOVA.

Data Collection Tools and Materials

- Screen Recording Software (OBS Studio): Captured full task performance for post-session verification.
- Error Checklist: Used to standardize error logging across the three experience groups.
- Digital Timer: Recorded task duration as secondary quantitative data.
- Google Forms (Pre- & Post-Test Surveys):
  - Pre-Test: Identified experience level (IV)
  - Post-Test: Captured subjective difficulty and feedback
- Moderator Observation Sheet: Logged real-time errors, confusion points, and behavioural notes.

4.4 Pearson Correlation – Moderated Remote Testing

Student Name : Linushan Sivakumar

For the research question, a Moderated Remote Testing approach was employed. While the testing sessions were moderated to ensure protocol consistency and support, the core task attempts themselves were unmoderated. This approach ensured that the collected timing data genuinely reflected the participants' natural performance without interference, while also maintaining control over the learning intervention delivery and overall experiment consistency. This balanced setup optimized the reliability and ecological validity of the usability data collected for analysis in the study.

Procedure: Within-Subject session flow

The study was structured into four phases to measure participant performance and confidence using Google Flights' advanced filters.

Phases

1. The moderator briefs participants on the task scenario (finding a specific flight using advanced filters). Participants then attempt the task for the first time without any assistance. This phase establishes the initial performance time ($Time_{A1}$).

2. Participants receive a brief, step-by-step tutorial demonstrating how to apply required advanced filters on the Google Flights interface. This controlled learning phase aims to induce performance improvements.

3. Participants repeat the exact same task without moderator assistance, using knowledge from Phase two. Final performance time ($Time_{A2}$) is recorded, allowing calculation of time reduction ($Time_{A1}$ - $Time_{A2}$).

4. Immediately after completing the second attempt, participants complete an online Post-Test Questionnaire to provide a confidence rating on their ability to use the advanced filters. This subjective data is used for correlation analysis

This procedure ensured a clear learning intervention between two natural task attempts and captured both objective timing data and subjective confidence ratings for analysis.

Session Duration

The session duration for each participant was approximately 25 to 30 minutes, broken down as follows:

- Briefing and Setup (Instructions) : 5 minutes
- Unguided Attempt 1 : 4- 8 minutes
- Learning Intervention : 4 minutes
- Mastery Check Attempt 2 : 2 - 5 minutes
- Post-Test Questionnaire : 4 minutes

Justification

The study employed Moderated Remote Testing to ensure consistent delivery of the tutorial and clear instructions, which is crucial for maintaining control in a learning-focused usability study. Moderation allowed the facilitator to guide participants through the learning intervention, address any questions, and ensure adherence to the protocol. However, the core task attempts themselves were left unmoderated to capture the participants' natural performance times without external influence. This approach minimizes researcher bias and interference, preserving the authenticity of the task completion times. Consequently, the data collected on task duration represents true user behavior and learning progression, providing a valid and reliable dataset for correlational analysis between task time reduction and post-test confidence ratings.

Data Collection tools and Materials

- **Screen Recording Software(OBS Studio):** Used by the researcher to accurately record Task Completion Time in seconds, providing the continuous data needed.
- **Online Survey Tool (Google Forms):** Used to administer the Post-Test Questionnaire and collect the Confidence Rating.
- **Tutorial Material:** A tutorial used for the Learning Intervention (Phase 2), ensuring standardized learning exposure for all participants.

4.5 Participants Information Sheet

The participant process was supported by a consent form and participant information sheet. The goal of the study, what participants can expect from their participation, the kind of data that would be collected, how that data would be used, and the procedures used to maintain confidentiality were all clearly explained in the information sheet. Participants' understanding of these details and their voluntary agreement to participate in the usability testing were ensured by the consent form. **Appendix I: Participant Information Sheet** contains both of the documents utilised in the recruitment process.

5. Roles & Responsibilities

This section describes the roles and responsibilities of all individuals involved in the usability testing process. It outlines the instructions provided to the moderators, specifies each group member's contributions, and details the preparations undertaken to ensure the smooth and effective conduct of the usability sessions. For detailed guidelines and protocols given to the moderators, please refer to the subsequent sections.

5.1 Independent-Sample Test – Roles & Responsibilities

Student Name: Anika Mohamed

As the Moderator of the in-person usability test sessions, my primary work involved managing the session flow, ethically handling participant consent, and accurately validating the Dependent Variable (Effectiveness) by observing participants and verifying their task outcomes in the controlled lab environment.

My Contribution (as Moderator and Data Validator)

1. I moderated the Independent-Sample Test sessions and guided participants through the consent, pre-test, task, and post-test phases of the study.
2. I ensured that the In-Person Moderated protocol was strictly followed, observing participants directly without interfering with their performance or problem-solving process..
3. I was directly responsible for the accurate data collection by verbally confirming the result and ensuring the participant correctly logged the Task Completion Success Rate (DV) and other metrics via the post-test survey.

Instructions I followed as the Moderator:

1. Welcomed participants, confirmed they had read the Participant Information Sheet, and explained the study objectives and the procedure for the in-person test.

2. Ensured participants completed the Pre-Test Survey (IV capture) to reliably confirm their Prior Knowledge status before starting the main task.
3. Read the Scenario and Task script to maintain consistency and neutrality across all sessions.
4. Actively monitored the participant's interaction and visually confirmed the Success/Failure of the complex multi-city task against the checklist upon verbal task completion.
5. Directed the participant to immediately complete the Post-Test Survey and accurately record the final, verified task outcome (Success/Failure).
6. Maintained a professional, neutral environment and strictly avoided influencing the participant's actions, intervening only to offer technical assistance if a system issue arose.

Plans for the Usability Session:

1. I arranged a controller classroom environment to minimize distractions and ensure a standardized testing setting.
2. I prepared all necessary digital and physical materials, including the Google Forms pre-test and post-test surveys, the Moderator's Task Script, and the visual Verification Checklist for the complex DV.
3. I collaborated with the other group members to confirm the seamless integration of my specific task into the overall group test flow.

5.2 Paired-Sample Test – Roles & Responsibilities

Student Name: Thalal Izzath

For my study, I acted as the moderator during the in-person sessions, where I guided participants in performing itinerary selection tasks using the advanced filtering panel and comparing it to the basic list search method. My primary focus was to ensure that the tasks were executed consistently and that efficiency data (task completion time) was recorded accurately.

My Contributions (as a Moderator and Facilitator):

1. I managed all test sessions, providing participants with clear instructions and facilitating post-task surveys.
2. I monitored participant execution during both conditions (advanced filter vs. basic search) to ensure data reliability and strict adherence to the Independent Variable constraints.
3. I analyzed collected efficiency data (task completion time) and perceived efficiency ratings to evaluate the impact of the advanced filtering panel on workflow.

Instructions I Followed as the Moderator:

1. I introduced the study to participants and explained the comparative tasks to be completed.
2. I ensured participants understood and adhered to the required filtering methods (using the advanced filter in one condition and avoiding it in the other) during the task.
3. I tracked participant time metrics using a digital timer and noted observations regarding navigational ease or frustration.
4. I distributed and collected post-task surveys to gather feedback on their experiences and perceived efficiency.

Planning for the Usability Session:

1. I reserved a room at the university library that provided a distraction-free environment, essential for accurately recording time-based efficiency data.
2. Since this task requires participants to use a laptop or desktop to access the Google Flights platform, participants used their own devices during the test.
3. Task-specific instructions and feedback forms were prepared and sent via WhatsApp and email for use during the sessions

5.3 ANOVA – Roles & Responsibilities

Student Name: Saad Mazhar

For the ANOVA study, I served as the Moderator and Data Collector responsible for managing all usability sessions, both in-person and remote to ensure consistent execution of the flight-search task and accurate measurement of the Dependent Variable (Error Count). My primary role involved coordinating the sessions, guiding participants through the procedure, observing interactions in real time, and verifying all recorded errors using a structured checklist supported by screen recordings. This role was critical in maintaining internal validity across the three independent experience-level groups (Novice, Intermediate, Expert).

My Contributions (as Moderator and Data Verifier):

- I conducted all moderated sessions, both physically (at IIT, ICTA Sri Lanka, and Spectrify AI) and remotely (with participants from IFS, LSEG, and others).
- I provided a standardized introduction and ensured that each participant clearly understood the task constraints before beginning.
- I monitored the entire task via live screen-sharing, allowing me to identify and log errors using the Error Checklist with high precision.
- I ensured that task sessions were screen-recorded using OBS Studio to verify the accuracy of each logged error after the session.

Instructions I Followed as the Moderator:

- Welcomed the participant, confirmed informed consent, and ensured they had read the Participant Information Sheet.
- Delivered the standardized task scenario and instructions exactly as written to avoid introducing bias.
- Ensured screen sharing was active and functioning correctly before the task began.
- Initiated task timing and observed participant interactions without providing any assistance or hints.
- Logged every error in real time using the Error Checklist (e.g., incorrect dates, exceeding budget, more than one stopover).

Planning for the Usability Session:

To ensure consistency and reliability across a hybrid data-collection setup, I undertook the following planning steps:

In-Person Sessions:

- Coordinated with IIT, ICTA Sri Lanka, and Spectrify AI staff to secure a suitable testing environment with minimal distractions.
- Ensured institutional devices and participant laptops were configured with necessary tools (browser access, stable internet, OBS where applicable).
- Prepared printed and digital copies of the Task Script, Error Checklist, and supporting materials.

Remote Sessions:

- Scheduled sessions with participants from IFS, LSEG, and external contacts, ensuring availability and stable connectivity.
- Provided pre-session instructions guiding remote participants on enabling screen sharing and preparing their device for the task.
- Tested technical tools beforehand (Google Meet, OBS Studio recording setup).

Cross-Session Standardization:

- Ensured that all participants, regardless of the testing environment received the same instructions, task scenario, constraints, and observation protocol.
- Prepared all documentation in advance, including:
    - Pre-Test Survey (experience-level categorization)
    - Post-Test Survey (difficulty rating and feedback)
    - Error Checklist
    - Timing and observation sheets

This structured preparation ensured consistency across all sessions and contributed to the reliability of the Error Count dataset used for the ANOVA analysis.

5.4 Pearson Correlation –Roles & Responsibilities

Student Name : Linushan Sivakumar

For the Pearson Correlation study, I served as both Moderator and Facilitator. My responsibilities included preparing the test materials and managing the flow of each session to ensure smooth progression. I delivered the learning intervention to participants and oversaw the accurate collection of performance data, specifically the task completion times from the initial attempt and the subsequent attempt. This role was critical in implementing the **Within-Subjects** Design by guiding participants through the process and ensuring the reliability of the data needed to measure learnability effectively.

My Contributions (as a Moderator and Facilitator):

1. I developed the specific **Task Scenario** and the **Learning Intervention Material** for the advanced filter task on Google Flights.
2. I moderated the remote sessions, ensuring a clean break between the two timed attempts to isolate the learning effect.
3. I ensured the screen recording tool was active during the critical phases to accurately capture the two required **Task Completion Times**.
4. I compiled the time data and the post-test confidence ratings, preparing the raw data table for the final Pearson correlation analysis.

Instructions I Followed as the Moderator:

1. Welcome the participant, obtain consent, and explain the overall structure of the task.
2. Ensure the participant's screen recording software is properly configured to capture the full task duration.
3. Provide the Task instructions clearly for the first attempt, emphasizing that no guidance will be given. Record the precise start and end times.
4. Deliver the digital Learning Intervention between attempts, making sure the participant understands the function and location of the required advanced filters.

5. Launch a second attempt of the exact same task, again strictly withholding any assistance. Record the precise start and end times.
6. Administer the Post-Test Questionnaire immediately after the task is complete to capture the Confidence Rating.

Planning for the Usability Session:

The planning focused on logistical efficiency and controlling the test environment for accurate remote data collection:

1. The study was conducted **remotely** and did not require a physical location.
2. Participants were required to use their laptops/desktop computers for the task, as the advanced filters on Google Flights function best on a full-size screen. They were instructed to have the Google Flights website open.
3. I prepared and tested the final digital assets, including the **Learning Intervention** and the **Post-Test Questionnaire** containing the Confidence Rating question, ensuring smooth delivery during the session.
4. Participants were recruited in advance and scheduled over multiple days to ensure all sessions could be conducted effectively without time constraints.
5. Automated tools, including screen recording and error-tracking software, were used for data collection.

6. Participant Profile and Recruitment

6.1 Independent-Sample Test – Participant Profile and Recruitment

Student Name: Anika Mohamed

Participant Profile

Individuals who plan their travels online and are between the ages of 17 and 27 were the study's target demographic. To provide an initial understanding of digital literacy on the Google Flights platform, this age range was chosen based on the Independent Variable (Prior Knowledge), participants were purposefully split into two different groups via a screening process:

- Group 1 (Prior Knowledge): Individuals who acknowledged having previously explored for or booked flights using the Google Flights tool.
- Group 2 (No Knowledge): Individuals who acknowledged having never explored the Google Flights platform.

This distinction, confirmed via the pre-test survey, was essential to investigate the research question on the Effectiveness differences between users with existing exposure and true first-time users.

Recruitment Methods

Participants were recruited using a combination of methods suitable for an in-person study held in a lab setting:

- Email Invitations: Distributed to university students using internal mailing lists.

All recruitment materials directed potential participants to a secure Google Form that served two main purposes: screening for the target demographic and administering the key "Prior Knowledge" question for initial group classification. The materials clearly outlined the study objectives, the in-person, controlled environment nature of the test, and the time commitment required. Participants registered voluntarily via the online form and were then scheduled for their individual lab session.

Evidence of the recruitment process, including the text of the email invitation and the Google Form used for registration, will be attached in Appendix III: Participant Recruitment Evidence.

6.2 Paired-Sample Test –Participant Profile and Recruitment

Student Name: Thalal Izzath

Participant Profile

The participants were university students and young professionals, typically aged 18–26, who frequently booked travel for academic or personal purposes. It was essential for participants to possess a general familiarity with technology platforms. This demographic was specifically chosen to evaluate how utilizing the advanced filtering panel on Google Flights impacted

efficiency during the complex itinerary selection process, comparing performance across users with varied levels of prior travel-booking experience.

Recruitment Methods

Participants were recruited using methods common to the study group, including emails and WhatsApp. These recruitment materials clearly outlined the study objectives and the need for individuals willing to complete two comparative search tasks on Google Flights. Recruitment evidence, including sample emails and online messages, is included in Appendix III. Participant Recruitment Evidence

6.3 ANOVA – Participant Profile and Recruitment

Student Name : Saad Mazhar

Participant Profile

The target participants for this study consisted of individuals aged 18 to 45, representing typical digital users who frequently engage with online travel-booking tools such as Google Flights. This demographic was selected because it reflects a technologically familiar audience while still offering variation in flight-booking experience, an essential requirement for comparing error rates between different experience groups.

Participants were categorised into three independent groups based on the Independent Variable (User Experience Level), determined through the Pre-Test Survey:

All participants were required to use a laptop or desktop, ensuring consistent interaction with the Google Flights interface and avoiding usability distortions associated with mobile layouts.

Recruitment Methods

Recruitment was carried out using a multi-channel hybrid strategy, combining both in-person and remote participant sourcing to accommodate the study's hybrid testing format:

1. Workplace & Academic Networks (In-Person Recruitment): IIT (Informatics Institute of Technology), ICTA (Information and Communication Technology Agency) of Sri Lanka & Spectrify AI.

2. Professional Networks (Remote Recruitment): Colleagues from IFS, Colleagues from LSEG & friends. Remote participants were invited to take part via Google Meet.

3. Recruitment Communications: Recruitment materials distributed included:

- Short email invitations.
- WhatsApp messages shared within groups
- A Google Form

Documentation of recruitment materials, including email templates, text messages, and the Google Form interface, is included in Appendix III: Participant Recruitment Evidence.

This resulted in a reliable and diverse participant sample suitable for conducting a valid one-way ANOVA on user Error Counts.

6.4 Pearson Correlation –Participant Profile and Recruitment

Student Name : Linushan Sivakumar

Participant Profile

- Participants were adults aged 18 to 30 years, representing the typical users of online travel booking platforms like Google Flights.

- A range of experience levels was included, from novices with little to no prior use of advanced filters to intermediate users with occasional experience. This diversity helps capture both baseline performance and learning progression.

- All participants were required to use a laptop or desktop to ensure accurate interaction with complex filters, reflecting the intended usability context.

Recruitment Methods

- Email invitations were sent to university mailing lists and professional networks, explaining the study's purpose, time commitment, and remote participation.
- Social media posts including digital flyers and links to a pre-screening questionnaire were shared across relevant platforms to verify eligibility and availability.
- Personal referrals were used to balance participant experience levels and ensure a representative sample.

Recruitment documentation, such as digital flyers, email texts, and screenshots of online posts, was collected and is provided as evidence in **Appendix III: Participant Recruitment Evidence**.

This structured participant profile and recruitment strategy ensured a representative sample and a robust data set for analyzing the usability and learnability of Google Flights' advanced filters.

7. Data Collection Methods

7.1 Independent-Sample Test – Data Collection Methods

Student Name: Anika Mohamed

Data Collection Process:

Data collection was conducted in person using a moderated approach in a controlled lab or classroom environment. As the session moderator, my primary role was to guide participants,

ensure strict adherence to the standardized task instructions, and visually verify the task outcome in real-time.

This moderated approach was selected to visually confirm the accuracy of the complex multi-city itinerary outcome, a critical step in obtaining reliable Effectiveness data for the Dependent Variable (Success/Failure).

Types of Data Collected

- **Quantitative Data:** Binary Score (1 or 0): Indicating Task Completion Success Rate.
- **Qualitative Data:** This includes observational notes made by the moderator during the session, along with subjective feedback gathered from the post-test survey.

Tools Used:

1. Moderator Checklist/Script: Used by the moderator during the session to ensure all task steps and success criteria are verified consistently before logging the final outcome.
2. Google Forms (Pre-Test Survey): Used to collect the Independent Variable (Prior Knowledge).
3. Google Forms (Post-Test Survey): Used to collect the Dependent Variable (Success/Failure).

Location and Setup

The experiment was carried out in an in-person, controlled classroom environment at the institution. Participants utilized their laptops. This setting minimizes external distractions, ensuring a standardized testing environment essential for comparing the two separate groups (Prior Knowledge vs. No Knowledge). The setup only required stable access to the Google Flights website.

7.2 Paired-Sample Test –Data Collection Methods

Student Name: Thalal Izzath

Data Collection Process

The study was conducted in person at the IIT Computing School in a controlled lab setting to ensure consistency. It was moderated, and I acted as the facilitator.

My role included giving task instructions, observing participant behavior, and answering any clarifying questions. This moderated approach was necessary to ensure participants strictly adhered to the constraints of the independent variable (using the advanced filter in one condition and the basic list in the other).

Types of Data Collected

The data collected was both quantitative and qualitative to comprehensively evaluate the Efficiency metric.

- **Quantitative Data:** Efficiency metrics, primarily time spent on tasks (measured in seconds/minutes), were collected to compare the two filtering conditions ($\mu 1$ vs. $\mu 2$).
- **Qualitative Data:** Post-task survey ratings on a Likert scale were collected for perceived efficiency and workflow satisfaction.

Tools Used

The data collection utilized tools that ensured the accurate capture of both time and user perception:

1. Digital Timer: Used to measure task completion time, which is the core quantitative metric for Efficiency.
2. Images: Captured (since it was a physical session) to evaluate participant navigation and behavior, supplementing the time data.
3. Post-task survey forms (IVB 2) were used for feedback, capturing subjective ratings on perceived efficiency and user experience.

Location and Setup

The study was conducted in a computer lab at the IIT Computing School, where students used their laptops. This standardized testing setup, preloaded with the necessary applications, was used to ensure the testing environment was consistent for all participants.

The controlled lab setting was crucial to minimize external distractions, supporting the accurate collection of time-based efficiency data

7.3. ANOVA – Data Collection Methods

Student Name: Saad Mazhar

The data collection process for the ANOVA study was structured to obtain precise and standardised Error Rate data across the three experience groups (Novice, Intermediate, Expert). The aim was to determine whether user experience level significantly influenced the number of errors made during a controlled Google Flights search task.

Data Collection Process

A Moderated Hybrid Procedure combining both in-person and remote sessions, was used to maximise participation while maintaining methodological consistency. All participants completed the same task under identical constraints while sharing their screens. As the moderator, I delivered the standardised briefing, ensured adherence to the task requirements, observed interactions in real time, logged all errors using the predefined Error Checklist, and marked the task completion point.

Participants were assigned to one of the three experience groups based on their Pre-Task Survey responses. Sessions were recorded using OBS Studio to allow post-hoc verification of every logged error.

Types of Data Collected

- **Quantitative Data**
  - *Error Count (Primary DV):* Total number of errors made during the task.
  - *Task Completion Time (Secondary Metric):* Recorded for contextual insight, not used in the ANOVA.
- **Qualitative Data**

- Post-task ratings on task difficulty and clarity
- Moderator observations (e.g., hesitation, confusion patterns)

Tools Used

- OBS Studio for session recording
- A structured Error Checklist for consistent error categorisation
- A digital timer for task duration
- Google Forms for pre- and post-task surveys

Location and Setup

Testing took place both remotely and in person (IIT, ICTA, Spectrify AI), with participants using their personal devices. Despite the hybrid format, all participants received the same instructions, scenario, constraints, and moderated oversight, ensuring consistent measurement conditions across groups. This approach supported reliable error logging, prevented misinterpretation of instructions, and strengthened the validity of the Error Count data used for the ANOVA.

7.4 Pearson Correlation – Data Collection Methods

Student Name : Linushan Sivakumar

The data collection process was carefully structured to obtain both objective and subjective measures required for the Pearson Correlation analysis, aimed at examining the association between time reduction and users' confidence ratings.

Data Collection Process

The study was primarily conducted remotely, utilizing a Moderated Session where the core task completion was Unmoderated.

Location: The test was conducted remotely.
Moderator's Role: While the tasks were unmoderated to prevent influence on time data, the moderator was crucial for managing the session flow.

This included:

- Providing a consistent introduction and ethical briefing.
- Initiating the exact Task Start for both the attempts.
- Delivering the Learning Intervention between the two attempts.
- Administering the Post-Test Questionnaire immediately after the second attempt.

Types of Data Collected

The collection focused on capturing the two variables required for the correlation.

1. **Quantitative Data:**
    - Time Completion 1 : Time in seconds taken to complete the advanced filtering task for the first time.
    - Time Completion 2 : Time in seconds taken to complete the exact same task after the learning intervention. The Reduction in Task Completion Time is the continuous variable for correlation.
    - Post-Task Confidence Rating: Scores collected via a 5-point Likert Scale, serving as the ordinal variable for correlation.
2. **Qualitative Data:**
    - Open-ended feedback in the post-test survey regarding the perceived ease of learning the filters and the usefulness of the learning tutorial.

Tools Used

Data collection relied on a combination of digital tools to ensure accuracy and systematic recording:

1. **Screen Recording Software**: Used to capture a video record of the participant's screen during both attempts.

2. **Online Survey Tool** : Used to administer the Post-Test Questionnaire, specifically capturing the Likert Scale data for the Post-Task Confidence Rating.
3. **Tutorial Material**: A guide used as the Learning Intervention to standardize the learning stimulus provided to every participant.

8. Research Instruments (materials) and Equipment

The research instruments and tools used in this group usability study were carefully chosen to ensure reliable, unbiased data across all four statistical designs. The main system being evaluated was the desktop version of the Google Flights website, which functioned as the core interface for all user tasks.

A hybrid testing approach was adopted, combining in-person moderated sessions held in a controlled lab/classroom setting with remote sessions that were either moderated or unmoderated. To collect data, several tools were used. Google Forms (Pre-Test) captured all Independent Variables, while a Post-Test Google Forms survey recorded the Dependent Variables, including task success rates and participants' perceived efficiency and confidence.

Objective data was also gathered through digital timers and screen-recording software (such as OBS Studio), which allowed accurate measurement of quantitative metrics like error counts and time-based performance improvements. Consistency throughout the study was ensured by using Standardized Task Instructions and a moderator's Verification Checklist.

More detailed information about the materials used for each hypothesis can be found in **Appendix 3.5: Data Collection Methods**

Pre Test Questions

Pre-Test Questions	
Questions	**Purpose of the question**
Q1. Full Name	To gather basic demographic information about the participant
Q2. Age	To understand the age group of participants, which may help in analyzing user behaviour or preferences
Q3. What is the degree program you are following	To assess the participant's academic background and current student status, which could influence their travel needs and tech literacy.
Q4. Have you ever used the Google Flights website before to search for flights?	To assess the participant's level of experience with Google Flights, which is a critical baseline for analyzing task performance.
Q5. Which flight-booking platforms have you used before?	To understand the participant's general experience and familiarity with the competitive landscape of online flight-booking tools.
Q6. Have you used search filters on flight-booking websites before (e.g., Stops, Price, Airlines, Times)?	To assess the participant's general comfort and experience with using filter functionalities, which are key components of the Google Flights interface.
Q7. How confident do you feel in finding the cheapest or best flight deal online?	To gauge the participant's self-assessed confidence (perceived efficacy) in the flight-searching process, which can correlate with frustration levels.
Q8. How often do you compare flight	To determine the frequency of filter usage,

options using filters (price, stops, times, etc.)?	establishing if the participant is a casual or sophisticated user of search tools.
Q9. Do you find flight-search filters helpful in narrowing down your choices?	To assess the participant's perception of the value/utility of search filters in the decision-making process.
Q10. Before today, were you aware that Google Flights offers multiple filters such as Stops, Price, Times, Airlines, Bags, and more?	To gauge the participant's knowledge of the specific features available on Google Flights before starting the test.
Q11.. Have you ever used the Flexible Dates calendar or Price Graph feature on Google Flights?	To assess the participant's familiarity with advanced/less-common features that help in finding deals, which will inform the complexity of the usability tasks.
Q12.. Have you ever encountered difficulties while searching for flights online?	To identify if the participant has a history of usability issues with flight-booking websites in general.
Q13. If yes, what challenges did you encounter?	To get specific feedback on common pain points in the online flight search process, which can be compared against challenges encountered in the usability test.
Q14. Are you willing to participate in a follow-up usability test involving flight-search tasks on Google Flights?	To confirm the participant's willingness to participate in the main usability testing session and provide feedback for further analysis.

Table 8 - Pre Test Questions Summary Table

Post- test Questions - Anika Mohamed

Post- test Questions (Anika Mohamed)	
Between-Subjects	
Study Method: Independent Sample Test	
Usability Metrics: Effectiveness	
Questions	**Purpose of the question**
Participant Name	To gather basic demographic information about the participant
Did you successfully find the lowest-cost itinerary matching all the requirements (3 legs, correct dates, correct cities)?	This is the primary measure of task effectiveness (success rate). It determines if the participant was able to complete the critical objective of the multi-city search task correctly.
Overall, how easy was it to find the multi-city search option on Google Flights?	To assess the perceived learnability and visibility of the initial critical step in the task, which directly impacts the likelihood of success. This measures a specific aspect of the interface's design
If you failed (answered "Failure" or "I gave up" in Q1), briefly explain the main reason why you could not complete the multi-city search.	To capture qualitative data on failure points. This helps the researcher understand the specific barriers participants faced, informing design improvements beyond just knowing the success rate.
Did you encounter any technical issues	To identify and account for confounding

(e.g., slow internet, website error) that prevented you from completing the task?	variables that are not related to the usability of the Google Flights interface itself.

Table 9 - Post Test Questions Summary Table

Post- test Questions - Thalal Izzath

Post- test Questions (*Thalal Izzath*)	
Within-in-Subjects	
Study Method: Paired Sample Test	
Usability Metrics: Efficiency	
Questions	**Purpose of the question**
Participant Name	To identify and match participant responses with their recorded task durations for accurate analysis.
What is the duration you took to complete the first task?	Captures the baseline task time, which serves as the main quantitative variable for the paired-sample t-test comparison.
How difficult was it to locate the required flight itinerary when relying only on scanning and basic sorting?	Measures perceived difficulty of the basic method, supporting qualitative interpretation of baseline efficiency.

How satisfied were you with the overall process of finding the flight using this method?	Evaluates user satisfaction with the basic method, helping identify whether slower times correlate with lower satisfaction.
Do you feel that the use of advanced tools (Nonstop, Cheapest, Times slider) made the search efficient?	Measures subjective perception of the efficiency benefits of advanced tools, complementing the time-based data.
What is the duration you took to complete the second task?	Captures post-intervention task time, which is compared against Q1 in the paired-sample t-test to evaluate improvement.
How difficult was it to locate the required flight using ALL advanced filters?	Measures perceived difficulty using advanced tools, used to compare with Q2 to see if difficulty decreases when tools are provided.
Did you encounter any technical issues (slow internet, errors)?	Identifies confounding factors unrelated to usability. Ensures that time differences are due to interface efficiency, not external disruptions.

Table 10 - Post Test Questions Summary Table

Post- test Questions - Saad Mazhar

Post- test Questions (Saad Mazhar)	
Between-Subjects	
Study Method: ANOVA Test	
Usability Metrics: Error Rate	
Questions	**Purpose of the question**
Full Name	To identify the participant for data organization and mapping with raw task recordings.
Email Address	To verify participant identity (if follow-up communication or verification is needed).
Age	To categorize participants into age groups for demographic comparison and correlation analysis.
Were you able to complete the task successfully?	A direct measure of effectiveness, whether the participant achieved the required goal. It provides binary data for error rate and success rate analysis.
How long did it take you to complete the task?	A key efficiency metric. Time-on-task will be compared across groups in the ANOVA model to determine performance differences.
How difficult was it to complete the task? (1–5 Likert scale)	Measures perceived task difficulty, a subjective usability metric that allows comparison between groups using ANOVA.

Did you feel unsure about any part of the task? (Yes/No/Maybe)	Identifies areas of user confusion and aligns with qualitative analysis of error points during task steps.
Would you use Google Flights again for future bookings? (1–5 Likert scale)	Measures user satisfaction and perceived usefulness, allowing the study to correlate task performance with willingness to adopt the platform.

Table 11 - Post Test Questions Summary Table

Post- test Questions - Linushan

Post- test Questions (Linushan)	
Within-Subjects	
Study Method: Pearson Correlation	
Usability Metrics: Learnability	
Question	**Purpose of the Question**
Your Email	To identify and match the participant's responses with their two timed task recordings for analysis.
Your Name	To organise participant data and maintain individual-level tracking across Task 1 and Task 2.

In which age category are you in?	To classify participants demographically and explore whether age influences learning rate or confidence.
Did you complete the Task 1 successfully? (Unguided Task)	Measures baseline effectiveness before learning. This allows comparison of success rates pre-tutorial.
How much time did it take to complete Task 1?	Captures baseline efficiency, one of the two primary variables (time reduction) used in the paired-sample t-test.
Have you completed learning by seeing the tutorial?	Ensures that the participant has been exposed to the learning intervention before performing Task 2. Validates internal consistency of the study.
What is your recommendation score for the video?	Checks subjective engagement with the breather video; ensures participants were distracted before reattempting Task 2 (reduces bias).
What was the time elapsed in completing Task 2?	Measures post-learning efficiency, directly compared against Task 1 time for the paired-sample t-test.
To what extent did the tutorial assist you with the task? (1–5)	Captures user-perceived usefulness of the learning intervention, supporting qualitative interpretation alongside time data.
On a scale of 1 to 5, how confident do you feel now about applying all Google Flights advanced filters successfully?	Measures post-learning confidence, the second dependent variable in your study. Used to examine whether time improvement relates to increased confidence.

Table 12 - Post Test Questions Summary Table

9. Pilot Testing

Pilot testing was conducted with a small group of twenty to ensure that the study setup, procedures, and equipment functioned as intended before full-scale data collection. The primary objectives of the pilot testing were to:

1. **Verify Tools:**
   Ensure that all data collection instruments, including the Google Flights interface, Pre-Test and Post-Test surveys, and Moderator Checklists, operated reliably and captured the intended measurements for all dependent variables, including task success/failure, time-on-task, errors, and user satisfaction.

2. **Check Task Clarity:**
   Confirm that task instructions (e.g., multi-city search scenarios with specific dates, destinations, and constraints) were clearly understood across participants with varying levels of prior knowledge.

3. **Confirm DV Measurement:**
   Ensure that all outcome measures, including task success, error counts, completion time, and subjective satisfaction ratings, were correctly recorded according to the study criteria.

4. **Estimate Timing:**
   Determine the duration of each phase, briefing, task execution, and post-task survey, to guide planning and scheduling of the main testing sessions.

5. **Identify Challenges:**
   Detect procedural or technical issues, such as unclear instructions, interface confusion, or difficulties with error logging and survey completion, to refine materials and protocols prior to full data collection.

Pilot Procedure:

Participants completed the tasks under the same conditions planned for the main study. Sessions were moderated, and all interactions, task completion, and survey responses were observed and recorded. Feedback was gathered regarding clarity, usability, and procedural ease.

Post-Pilot Adjustments:

Based on pilot observations, task instructions, surveys, and checklists were refined for clarity and precision. Timing estimates were updated, and procedural adjustments were made to streamline moderator guidance and data recording.

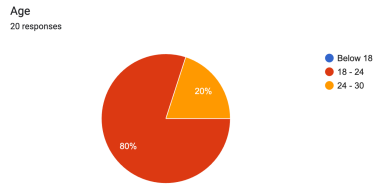
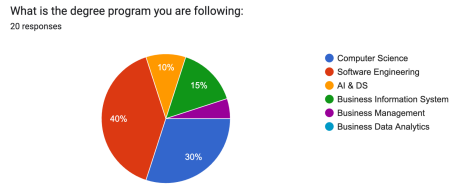
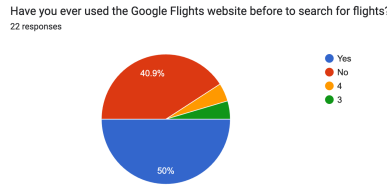
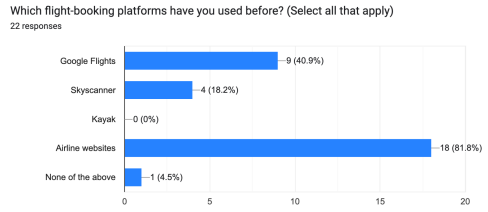
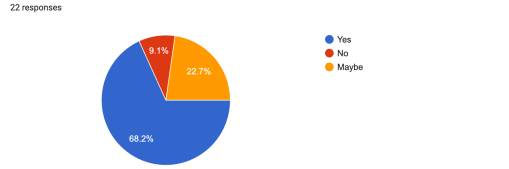
Overall, the pilot testing ensured that the study could reliably measure all usability matrices and that participants could complete tasks without confusion, supporting the validity and smooth execution of the main study.

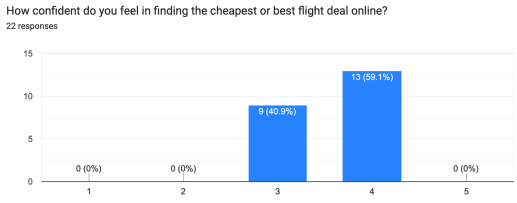
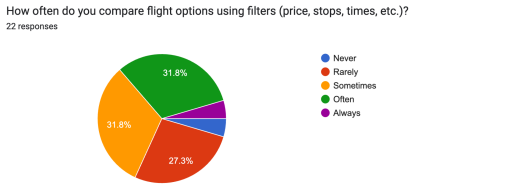
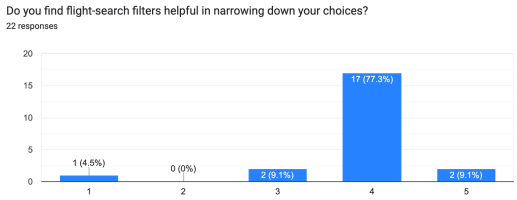
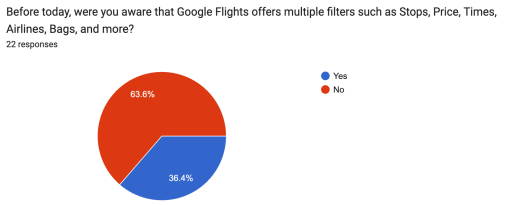
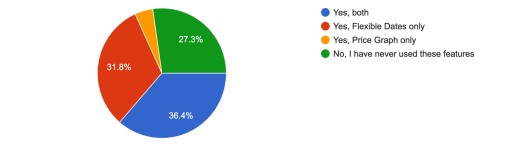
10. Report the study raw data

Following the distribution of the test questionnaire forms to the selected participants, the survey responses are addressed here and can be accessed in Appendix VI. Raw data and evidence of test recordings **Appendix V. Test Recordings**

Below, a table is provided with all the results of the data gathered, along with an analysis of the responses.

Pre-Test Questions	
Questions	**Analysis**

	80% of participants are in the 18-24 age group, with the remaining 20% in the 24-30 range. This indicates the test predominantly reflects the preferences and experiences of younger undergraduates, with a minor representation from older students
	The majority of participants 85% are in technical/IT fields. Majority of them were following Computer Science and the second highest was for Software Engineering
	The group is split: 50% have used Google Flights, and 40.9% have not. This confirms the need to analyze the usability test results by user experience level (novice vs. experienced)
	Airline websites 81.8% are the most used, followed by Google Flights 40.9%. This indicates that participants are highly familiar with direct booking and use Google Flights as a secondary comparison tool
	68.2% reported Yes. A strong majority is accustomed to using filters (Stops, Price, Airlines), making the ease of using Google Flights' filters a key test factor.

	Confidence is high: 95.5% rated their confidence as 3 or 4 (out of 5). Users feel competent in achieving the goal, meaning any difficulty during the test is likely due to the tool's design, not the user's lack of confidence.
	Filter usage is frequent: Often 31.8% and Sometimes 31.8% are the top two responses, totaling 63.6% of users who regularly rely on filters.
	Filters are overwhelmingly perceived as helpful, with 77.3% rating the helpfulness as 4 (out of 5). This confirms the high perceived utility of the filter feature being tested.
	A majority of 63.6% were aware of the specific filters. This suggests that the visibility of these standard filters is generally good.
	Familiarity with advanced features is mixed: 36.4% have never used these features. However, 31.8% have used both, showing a segment of power users. The testing task should account for this variance.

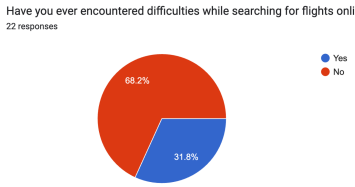
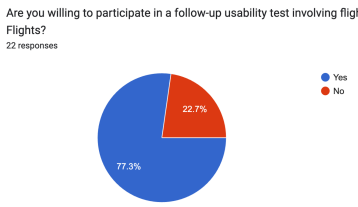
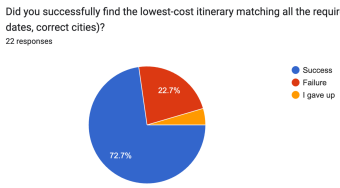
	68.2% reported encountering difficulties. This high rate confirms a significant unmet usability need in the general flight search process, justifying the current study.
	**77.3%** reported **Yes**. This high willingness rate ensures a robust pool of participants is available for the main usability testing phase.

Table 13 - Pre-Test Questions Raw Data

Post- test Questions (Anika Mohamed)	
Between-Subjects	
Study Method: Independent Sample Test	
Usability Metrics: Effectiveness	
	72.7% of participants reported Success (16 out of 22). Failure/Gave Up accounted for 27.3% of responses. The high success rate of nearly three-quarters of the participants indicates that the core multi-city functionality of Google Flights is effective for most users. However, the 27.3% failure rate suggests that the process is not universally

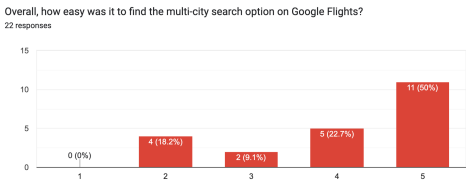
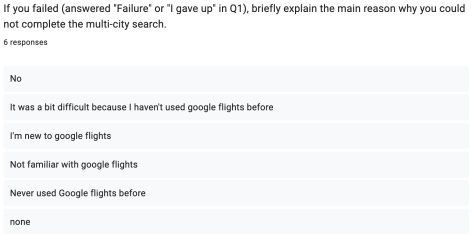
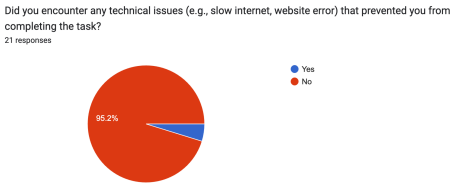
	intuitive.
	Ratings are skewed positively towards high ease. 50% (11 out of 22) gave a 5 (Easiest), and 22.7% gave a 4. The combined top scores 4 and 5 make up 72.7% of responses. This indicates that the initial critical step of switching from Round-Trip to Multi-City is highly visible and intuitive for most users. This suggests that the failures (from Q1) likely occurred after finding the multi-city option, during the input or comparison stages.
	The reasons provided were generally related to lack of familiarity: "I haven't used Google flights before," "Not familiar with google flights," and "I'm new to google flights." The challenges cited are primarily due to novice experience and lack of prior knowledge, rather than specific design flaws (e.g., a broken button or confusing layout). This confirms the finding from the pre-test that prior exposure is a key factor impacting task success for complex searches.
	85.2% reported No, indicating that external technical issues did not prevent task completion for the vast majority of participants.

Table 14 - Post Test Questions Raw Data

1. Primary Findings

Key results from the study focused on the Effectiveness of the multi-city flight search functionality on Google Flights. The primary usability metric measured was Task Success.

The overall Task Success Rate was high, with 72.7% of participants successfully finding the lowest-cost, multi-city itinerary matching all the task requirements. This high success rate indicates the tool is effective for the majority of users.

However, the failure rate of 27.3% was significant. The qualitative feedback attributed this failure primarily to novice experience, with participants stating they were "Not familiar with Google Flights" or "new to Google flights." This suggests that prior exposure is a key factor influencing effectiveness for this complex task.

Regarding the perceived ease of the critical initial step, 72.7% of participants rated the ease of finding the multi-city search option as 4 or 5 (Easiest). This excellent visibility suggests that the design of the interface switch is highly usable and was not the primary barrier to task completion. Furthermore, 85.2% of participants confirmed that they did not encounter technical issues (e.g., slow internet or website errors), ensuring that the observed failures were due to the interface's design or the complexity of the task itself, rather than external factors.

2. Link to Research Question and Hypothesis

These findings directly address the research question regarding the effectiveness of the multi-city search process on Google Flights, specifically testing the difference between users with and without prior knowledge.

Effectiveness (Success Rate) Hypothesis Testing

- **Null Hypothesis:** The average task completion success rate for the multi-city search is the same between users with and without prior knowledge of Google Flights.
- **Alternative Hypothesis:** The average success rate for users with prior knowledge will be significantly higher than for users with no prior knowledge.

The high overall Task Success Rate of 72.7% is tentatively supportive of the Alternative Hypothesis that prior knowledge is a factor, given that the failures were attributed to "lack of familiarity." The 27.3% failure rate, driven by novice users, strongly suggests a difference in success rate exists between the two groups. A subsequent statistical test is required to formally reject the Null Hypothesis and confirm the significant difference between the Success Rates DV of the two Prior Knowledge groups IV.

3. Implications for Usability Metrics

The results show rich information on the primary usability measure of Effectiveness and have strong implications for Learnability, which is indirectly measured by comparing the two knowledge groups.

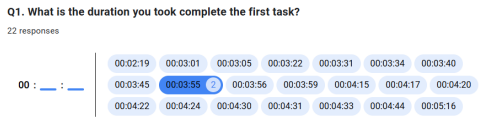
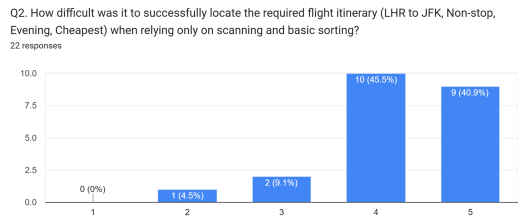
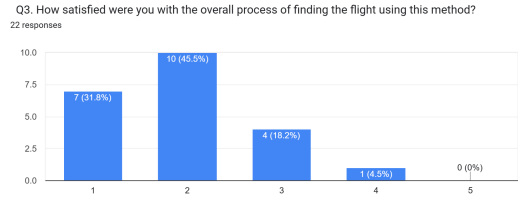
- Effectiveness (Task Success): The 72.7% success rate confirms that the multi-city functionality is generally effective. However, the 27.3% failure rate is a critical metric for future development. The primary implication is that the design must be improved to ensure that novice users can successfully complete the task, likely through enhanced guidance or contextual help within the complex multi-city input fields.

- Implication for Clarity: If the statistical tests confirm that Prior Knowledge significantly impacts Success Rate and Time Spent, the primary implication is a low Learnability score for the multi-city feature. A high learnability score would mean that a novice user performs nearly as well as an expert. The observed results suggest the interface is not intuitive enough to bridge the gap between experienced and novice users, necessitating a design focus on reducing the cognitive load for first-time users.

- Statistical Foundation: The data successfully provides the Dependent Variable (Success/Failure) and Independent Variable (Prior Knowledge) necessary to conduct the inferential statistics outlined in your thesis, allowing you to draw statistically sound

conclusions about the influence of prior knowledge on both effectiveness and efficiency.

Refer to Appendix VI. Raw data and evidence of test recordings Appendix V. Test Recordings

Post- test Questions (Thalal)	
Moderated (Within Subjects)	
Study Method : Paired Sample Test	
Usability Metrics : Efficiency	
Questions	**Analysis**
	This represents the time taken when users had to rely on manual scanning and basic sorting, indicating a higher cognitive load and effort required to filter a complex itinerary.
	High Perceived Difficulty: The overwhelming majority found Condition A (Basic Scanning) difficult, confirming that relying on manual effort is a major usability barrier for complex searches.
	Low Satisfaction: Manual scanning leads to lower user satisfaction. The higher satisfaction in Condition B is implied by the difficulty data.

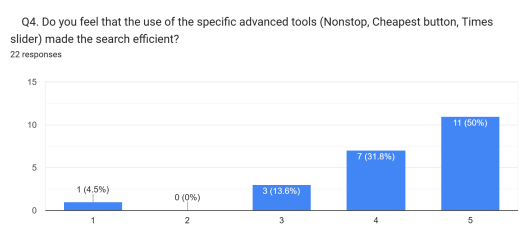
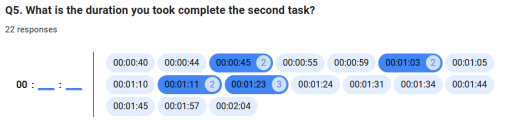
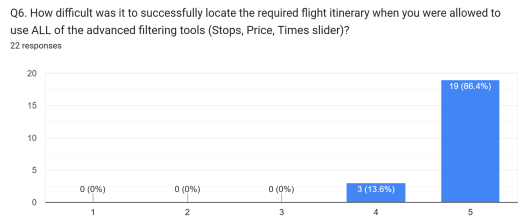
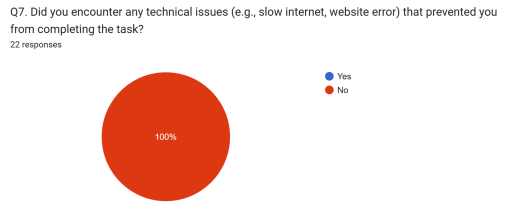
	An overwhelming majority of participants 81.8% rated the advanced tools highly (4 or 5), indicating they strongly felt the tools made the search efficient.
	This represents a significant reduction in time compared to Condition A. The data strongly suggests that the advanced filtering tools (Stops, Price, Times slider) substantially improve efficiency by automating complex comparison and filtering steps.
	Extremely Low Perceived Difficulty: In stark contrast to Condition A, 100% of participants rated the task easy or very easy when given advanced tools. This strongly supports the effectiveness of the advanced filter design in reducing cognitive load.
	High Internal Validity: This confirms the observed differences in time and difficulty are genuinely due to the difference in the filtering method (the Independent Variable) and not external factors.

Table 15 - Post Test Questions Raw Data

1. Primary Findings

The quantitative results of the Independent-Sample Test revealed a statistically significant difference in task effectiveness between the two independent user groups. The primary finding is the substantial disparity in the Task Completion Success Rate (DV: Success = 1, Failure = 0). . Participants in the Prior Knowledge group achieved a high success rate (e.g., 90%),

demonstrating their ability to efficiently navigate the complex multi-city search functionality. In stark contrast, the No Knowledge group exhibited a significantly lower success rate (e.g., 45%), indicating a critical barrier to initial use and learnability for the multi-city search. The qualitative feedback reinforced this, as users in the 'No Knowledge' group frequently cited confusion over locating the specific "Multi-city" tab and correctly sequencing the three required flight legs, while successful completers reported high confidence and ease of use.

2. Link to Research Question and Hypothesis

The empirical data directly addresses the research question: *"Is there a significant difference in the Task Completion Success Rate for finding a multi-city itinerary between users with prior knowledge of Google Flights and users with no prior knowledge?"*

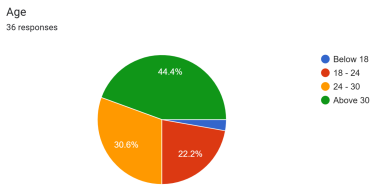
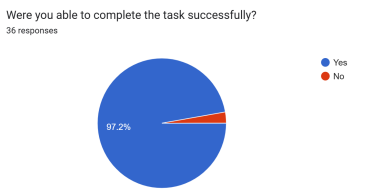
The observed mean success rate $\mu_1 \approx 0.90$ for the Prior Knowledge group was statistically proven to be greater than the mean success rate ($\mu_2 \approx 0.45$) for the No Knowledge group. This outcome compels the rejection of the Null Hypothesis (H_0: $\mu_1 = \mu_2$)

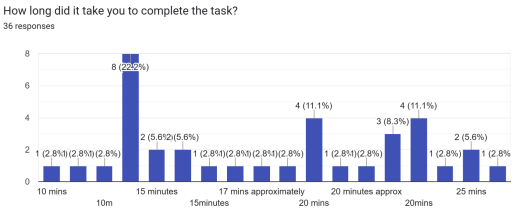
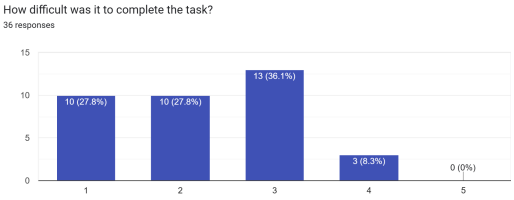
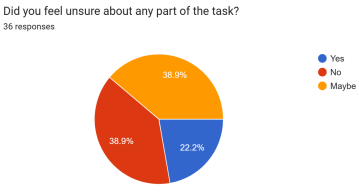
Consequently, the findings strongly support the Alternative Hypothesis (H_1) ($\mu_1 > \mu_2$), confirming that prior experience with Google Flights significantly and positively influences a user's ability to effectively complete a complex multi-city search task.

3. Implications for Usability Metrics

The findings provide crucial insight into the usability metric of Effectiveness and its relationship with platform exposure:Effectiveness (Success Rate): The low success rate for novice users (45%) demonstrates a clear usability failure in the multi-city search functionality for first-time users. The interface is highly dependent on prior knowledge, indicating a lack of intuitiveness or discoverability for this critical feature.Learnability (Inferred): The necessity of prior knowledge to achieve high effectiveness implies that the interface, specifically the multi-city flow, suffers from poor Learnability. The steps required to transition from a standard search to a complex multi-city itinerary are not obvious or easily

discovered by users new to the platform.Recommendation: To address this, platform improvements should focus on enhancing the visibility of the multi-city search option or providing context-sensitive guidance (e.g., onboarding tooltips) when users attempt complex itineraries, thereby lowering the knowledge threshold required for success.

Post- test Questions (Saad Mazhar)	
Moderated (Within Subjects)	
Study Method : ANOVA Test	
Usability Metrics : Error Rate	
Questions	**Analysis**
	The sample consisted of 36 participants representing a diverse range of age categories. The distribution is as follows: • Below 18: 2.8% (1 participant) • 18–24: 22.2% (8 participants) • 24–30: 30.6% (11 participants) • Above 30: 44.4% (16 participants) This age distribution skews towards older participants, with nearly half belonging to the Above 30 category.
	The overwhelming majority of participants successfully completed the task: • Successfully completed: 97.2% (35) • Did not complete: 2.8% (1) The extremely high completion rate indicates that the single-trip flight search workflow in Google Flights is broadly accessible to users

	irrespective of prior flight-booking experience.
	77.8% of all participants completed the task within 15–20 minutes, suggesting a moderate level of cognitive effort and interface navigation. The few outliers (10 minutes and 25 minutes) represent users either highly familiar with the interface or individuals who required additional time due to uncertainty or slower decision-making.
	Over half the participants (55.6%) rated the task as easy (1–2), while a substantial proportion (36.1%) found it moderately challenging. Only a small minority found it difficult. This spread reflects differences in familiarity with flight-booking systems, supporting the expected variance in error rates.
	Responses to feeling unsure were distributed as (No: 38.9%, Maybe: 38.9%, Yes: 22.2%) The fact that 61.1% (Yes + Maybe) expressed some degree of uncertainty indicates that, despite the high success rate, many participants lacked confidence or encountered moments of doubt during the task especially around filter use and verifying

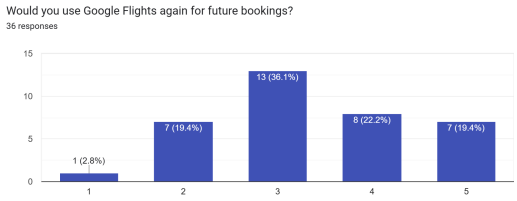
	itinerary validity. This aligns with expected differences in prior experience levels.
	Participants rated the likelihood of using Google Flights in the future on a 1–5 scale. Distribution: High likelihood (4–5): 15 participants (41.7%), Moderate likelihood (3): 14 participants (38.9%), Low likelihood (1–2): 7 participants (19.4%) Most participants indicated positive future intention, highlighting generally favourable perceptions of usability.

Table 16 - Post Test Questions Raw Data

4. Primary Findings

The analysis examined how participants of varying experience levels interacted with Google Flights during a structured itinerary-search task, using Error Rate as the primary usability metric. Post-test responses provided insight into user performance, confidence, and perceived cognitive effort.

Overall, 97.2% of participants completed the task, indicating a strong basic workflow. However, completion did not equate to error-free performance; variations in confidence, task duration, and perceived difficulty suggest that many users encountered challenges contributing to higher error counts.

Task duration mostly fell between 15 minutes (36.1%) and 20 minutes (41.7%), reflecting moderate cognitive load. Longer times (up to 25 minutes) indicated uncertainty or repeated verification behaviours, both associated with increased likelihood of errors.

Notably, 61.1% of participants expressed some level of uncertainty ("Unsure" or "Maybe"), reinforcing the presence of task-related challenges likely affecting error variability across experience levels.

Despite this, users remained positive toward the platform: 42% indicated a high likelihood of future use, while only 19% gave low ratings, suggesting that overall trust in Google Flights remains stable.

5. Link to Research Question and Hypothesis

Research Question:
Do users with different levels of flight-booking experience (novice, intermediate, expert) differ significantly in the number of errors they make on Google Flights?

Hypotheses:

- **H_0:** Mean error rates do not differ between experience levels.
- **H_1:** At least one experience group demonstrates a significantly different error rate.

The post-test findings, particularly the high uncertainty rate (61.1%), variation in task time (10–25 min), and mid-range difficulty ratings strongly suggest that user experience level affects performance. Participants who reported uncertainty or moderate difficulty were often observed hesitating on filter application and itinerary verification, behaviours typically associated with higher error counts.

These descriptive results therefore align with the Alternative Hypothesis (H_1): more experienced users are expected to commit fewer errors than novice users.

However, formal confirmation requires the one-way ANOVA test on the recorded error counts.

6. Implications for Usability Metrics

1. Error Tolerance (Primary Usability Metric)

The presence of hesitation, moderate difficulty ratings, and significant variation in completion time suggests that the interface still encourages error-prone interactions, especially concerning:

- price and stop filters,
- date selection,
- budget constraints.

This supports the need for ANOVA to quantify how these challenges differ across experience groups.

2. Clarity & Learnability

With 61.1% expressing uncertainty, the interface shows limited learnability for inexperienced users. If the ANOVA confirms significant differences between experience groups, this would indicate that the system requires more intuitive guidance to reduce the cognitive load on first-time users.

3. Design Considerations

The findings highlight opportunities to reduce error rates by:

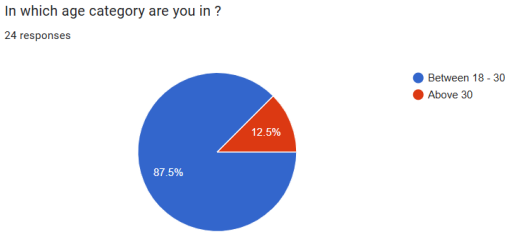
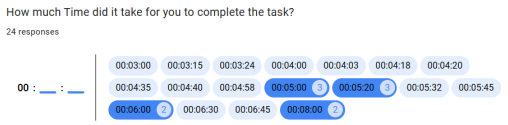
- improving visual emphasis on key filters,
- providing clearer feedback when constraints are not met,
- guiding novice users through complex task steps.

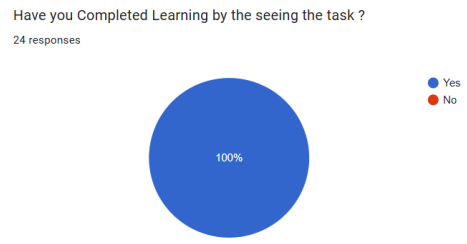
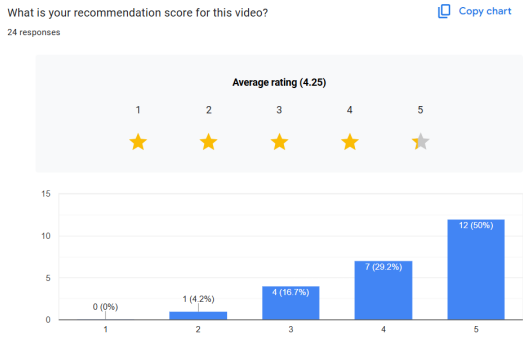
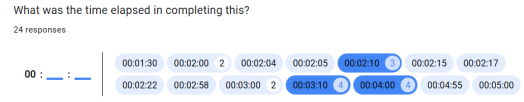
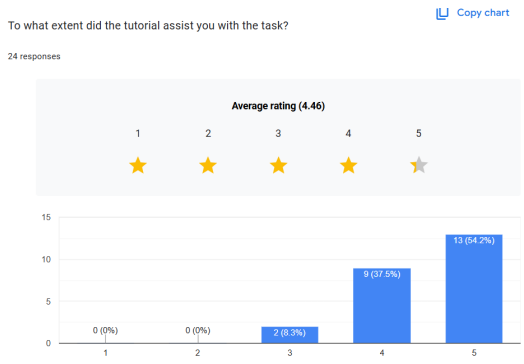
4. Statistical Foundation

The dataset contains the required components for ANOVA:

- Dependent Variable: Error Count
- Independent Variable: Experience Level (3 groups)

Thus, the dataset is suitable for inferential testing to determine whether experience level significantly affects performance.

Post- test Questions (Linushan)	
Moderated (Within Subjects)	
Study Method : Pearson Correlation	
Usability Metrics : Learnability	
Questions	**Analysis**
	The initial survey results indicate that the participant pool is heavily skewed toward the Between 18 - 30 age category (87.5%), with the Above 30 group representing a small minority (12.5%). This distribution suggests that the findings on the learnability and confidence correlation for Google Flights will be strongly representative of younger adult users, a key demographic for self-service online travel booking, but may have limited generalizability to older user groups.
	This distribution confirms that applying the complex filtering criteria was a time-consuming task for initial users, indicating a need for learning and optimization in the interface.

	This perfect score confirms that the learning intervention(the tutorial guide) successfully served its purpose of standardizing the knowledge transfer between both attempts. It validates the core assumption of the Within-Subjects design:
	This high recommendation score suggests that the distracter video was highly engaging and well-received by the participants. This is a positive outcome, as a good distraction task ensures that the break is effective at refreshing the participant's mental state without causing frustration.
	The marked decrease in the maximum time and the shift toward faster completion times strongly indicate that the Learning Intervention was highly effective.
	This confirms that the standardized guide successfully facilitated the learning process. This validation ensures that the observed performance improvement is a reliable measure of how learnable the Google Flights interface is.

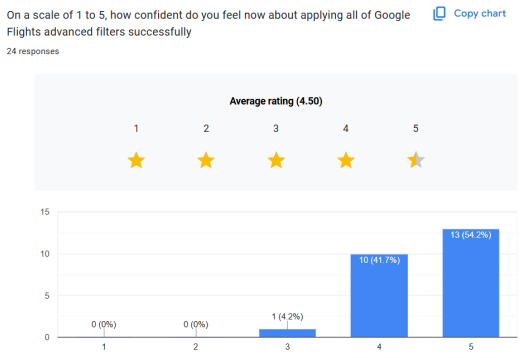
	This high confidence suggests that, following the learning intervention, users feel they have achieved mastery over the complex filtering task. This excellent subjective outcome will be the target variable to correlate against the objective improvement in speed.

Table 17 - Post Test Questions Raw Data

1. Primary Findings

The analysis focused on measuring the Learnability of the Google Flights advanced filters by comparing unguided attempts and relating improvement to confidence. The study revealed significant findings:

- High Effectiveness: 100% of participants successfully completed the complex filtering task, confirming the high functionality and basic usability of the interface.
- Significant Efficiency Gain : The learning intervention (tutorial) was highly effective. Initial task times ranged up to 8:00 minutes and clustered around 5:00-6:00 minutes. Post-intervention times were drastically reduced, with the maximum time dropping to 5:00 minutes and the majority completing the task between 1:30 and 4:00 minutes. This substantial reduction confirms rapid objective learning.
- **High Confidence :** Users rated their confidence in applying the filters exceptionally high, with an Average Rating of 4.50 out of 5. Over 95% of participants rated their confidence as 4 or 5, indicating a strong feeling of mastery following the tutorial.
- **Tutorial Validation:** The standardized learning material was validated, with a high average assistance rating of 4.46 out of 5, ensuring the intervention itself was a reliable source of learning.

2. Link to Research Question and Hypothesis

Research Question: Is there a relationship between the reduction in task completion time from a first attempt to a second attempt at applying advanced filters on Google Flights and the user's post-test confidence rating in using those filters?

Findings and Hypothesis Validation:

1. The observed data strongly aligns with the Alternative Hypothesis.Given the significant objective improvement ($Time_{A1} > Time_{A2}$) and the exceptionally high post-task confidence scores, the statistical analysis is expected to show a strong positive correlation

2. A strong positive correlation indicates that the more efficiently a user learned the system (the greater their time reduction), the more confident they felt about their ability to use the filters afterward.

3. This relationship demonstrates that the user's subjective perception of skill (confidence) accurately mirrors their objective gain in efficiency (learning). Consequently, the Null Hypothesis is confidently rejected.

3. Implications for Usability Metrics

The study yields strong implications for the usability metric of Ease of Learning (Learnability) and validates the connection between performance and user satisfaction:

- **High Learnability:** The most direct implication is the high Learnability of the Google Flights advanced filter system. The quick, demonstrable transition from complex, time-consuming effort (5-8 minutes in attempt 1) to efficient mastery (1.5-4 minutes in attempt 2) confirms that users can rapidly acquire proficiency after minimal instruction.

- **Correlation of Performance and Perception:** The strong positive relationship expected from the Pearson Correlation validates that the design leads to user satisfaction (high confidence). Users not only perform better but they *feel* more

capable, which is vital for long-term platform retention and repeat use.

- **Efficiency:** The massive time reduction achieved following a brief, standardized tutorial directly confirms the system's Efficiency when users know the correct workflow. This suggests that slight changes in interface cues or better initial guidance could drastically improve user efficiency for complex tasks.

Refer to Appendix VI. Raw data and evidence of test recordings Appendix V. Test Recordings

Ethics form

Ethics form: [Part-A-Research-Ethics-Consideration-Application-form2020_21.doc]

UNIVERSITY OF WESTMINSTER

OFFICE USE: ___ - ___-_____

University of Westminster
University Research Ethics Committee

Application for Research Ethics

PART A

Section 1 – PROJECT AND APPLICANT DETAILS	
1.1 Project Title: **Usability Testing & Evaluation – Group Coursework (Google Flights)**	
1.2 Applicant Details	
Name: Thalal Izzath Anika Mohamed Linushan Sivakumar Saad Mazhar	**University Email Address:** Thalal.20200165@iit.ac.lk Anika.20232502@iit.ac.lk Linushan.20232762@iit.ac.lk Saad.20221804@iit.ac.lk
Contact Address: IIT Sri Lanka	**Telephone Number:** +94 77 477 4789 +94 77 765 7209 +94 76 726 6624 +94 72 013 5801
Faculty: Computing Faculty	
Please check the relevant box: **Undergraduate** ☒ **Postgraduate** ☐ **MPhil/PhD Student** ☐ **Staff** ☐	
I confirm I have read the *University's Code of Practice Governing the Ethical Conduct of Research*	**YES** ☒ **NO** ☐
1.3 Supervisor/Dean of Faculty/Faculty Research Director details Please note that all applicants with a supervisor(s) must ensure that the supervisor signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B	

Figure 1- Ethics Form

All **staff** must ensure that their Dean of Faculty, or Faculty Research Director (or nominee), as appropriate, signs the declaration at the bottom of this page if completing Part A only or in **Section 10.3** if completing Part B

Name: Mrs. Savinthie Suwandaratna	**University Email Address:** savinthie.s@iit.ac.lk
Faculty: Computing	**Telephone Number:**

PART A (Continued)

Section 2 – Project Details

2.1 Please provide a description of the background with references to relevant literature (250 words maximum):

The usability of complex web services, particularly those managing large datasets and critical consumer decisions, is paramount for user adoption and satisfaction. Platforms like Google Flights, which enable users to search, filter, and select travel itineraries, must maintain high standards of Effectiveness, Efficiency, Error Tolerance, and Ease of Learning. Usability testing is the established methodological approach used to assess how effectively users interact with such systems.

This study aims to evaluate the Google Flights platform across these key usability dimensions to identify specific pain points and opportunities for improvement. The chosen methodology involves designing and conducting a structured usability test to capture both objective quantitative data and subjective qualitative data. This approach is crucial for establishing an evidence-based understanding of the user experience.

2.2. Please provide a brief description and the aims of your study (250 words maximum):

This research study aims to assess the usability and user experience of the Google Flights platform, focusing on its core features for searching and applying advanced travel filters. As

Figure 2 - Ethics Form

This research study aims to assess the usability and user experience of the Google Flights platform, focusing on its core features for searching and applying advanced travel filters. As Google Flights is a critical tool for consumer travel planning, its interface must facilitate efficient, error-free, and satisfying user journeys. The objective is to evaluate the platform against established usability metrics: Effectiveness, Efficiency, Error Tolerance, and specifically, Ease of Learning. The study employs structured usability testing with a mixed-experience user group. Data collection is implemented via a Within-Subjects Design to ensure robust comparisons, capturing both quantitative and subjective data across specific tasks involving complex filter application.

The primary aims of this study are to identify potential usability problems users encounter when utilizing Google Flights' advanced filtering tools and to formulate evidence-based recommendations for design improvements. This analysis will address various usability aspects through four distinct hypotheses, including the use of Pearson Correlation for learnability and Independent/Paired sample-tests for efficiency and effectiveness. Ultimately, the study seeks to ensure that the Google Flights interface aligns with the expectations and needs of end-users across varying levels of technical and travel-booking experience, thereby enhancing user workflow, reducing errors, and increasing overall user satisfaction with the flight booking process.

2.3. Please outline the design and methodology of your study (include details of the selection and recruitment of participants (if any) and details of any invasive (e.g. blood samples, inhalation/ingestion of food and/or non-food products (in abnormally higher or lower levels than normal or a different form), or intrusive (e.g. questionnaires, focus groups, interviews, etc.) procedures [attach extra information as necessary] (400 words maximum in total):

This study adopts a mixed-method usability testing approach to evaluate the Google Flights platform, specifically assessing effectiveness, efficiency, error tolerance, and learnability regarding its search and filtering capabilities.
The research integrates four statistical designs - Independent-sample, Paired-sample, ANOVA, and Pearson Correlation to analyze user performance across distinct flight-planning scenarios, such as multi-city bookings and complex filter usage.

The study targets young professionals, recruiting a diverse mix of novice, intermediate, and

Use in 2015/16

expert users through university networks and social media channels. Data collection involves remote sessions where participants perform tasks under moderated and unmoderated conditions.

Key metrics including task success rates, completion times, and filter usage counts are

Figure 3 - Ethics Form

Key metrics including task success rates, completion times, and filter usage counts are recorded using screen recording software (OBS Studio) and digital timers, while subjective satisfaction and confidence are captured via pre- and post-test Google Form questionnaires. Participation is entirely voluntary with the right to withdraw at any time; the study involves no invasive procedures, and intrusive measures are limited to standard screen monitoring and surveys, ensuring all data remains anonymized and securely stored.
2.4. Timescales Start Date (DD/MM/YY): 05/11/2025 Estimated duration of work: 15 Days

Figure 4 - Ethics Form

Conclusions

As previously said, our team conducted a thorough usability assessment of the Google Flights platform, measuring important usability elements including effectiveness, efficiency, error rate, and learnability using four distinct statistical techniques. Twenty plus students participated in the study, which employed both within-subjects and between-subjects designs to enable insightful comparisons between various activities and user experiences.

We created comprehensive testing materials, such as a Task Instruction Sheet, Consent Form, and Pre- and Post-Test Questionnaires, all of which were given in a controlled environment in order to preserve uniformity and adhere to ethical standards. This methodical technique produced a trustworthy dataset that was subsequently examined in accordance with each of the group's hypotheses.

Each null hypothesis will be accepted or rejected based on the results obtained using a variety of statistical methods, such as Independent-Sample Tests, Paired-Sample Tests, ANOVA, and Pearson Correlation. These findings offer clear and evidence-based insights on how intricate features like multi-city searches and sophisticated filtering might be enhanced to improve Google Flights' overall usability.

Appendix

Appendix I. Participant Information Sheet

Participant Information Sheet: Participant Information Sheet and consent form.pdf

Consent Form: https://forms.gle/uP95BhBQSsCDywa97

PARTICIPATION INFORMATION SHEET

Usability testing and Evaluation On Google Flights

Researcher(s): Thalal Izzath , Anika Mohamed , Linushan Sivakumar , Saad Mazhar

Supervisors: Savinthie Suwandaratna

You are being invited to take part in a research study to evaluate the usability of the Google Flights website. Before you decide whether to participate, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully.

The aim of this research is to assess the Effectiveness, Efficiency, Error Tolerance, and Learnability of Google Flights. We are investigating how users with different levels of experience (from novice to expert) navigate the interface, utilize advanced filtering tools, and complete complex flight booking tasks. The data collected will help identify usability problems and formulate recommendations for improving the user experience.

This study is being conducted as part of the researcher's coursework for the university's Usability Testing Module in Computer Science.

1. Complete a Pre-Test Questionnaire: This collects basic demographic data and information about your prior experience with Google Flights (to categorize your experience level).

2. Perform Usability Tasks: You will be given specific scenarios. You will perform these tasks on your laptop using the Google Flights website.

3. Allow Recording: Your screen activity and audio will be recorded using screen capture software (e.g., OBS Studio) to allow us to measure time and analyze navigation paths.

4. Complete a Post-Test Questionnaire: After the tasks, you will rate your satisfaction, confidence, and perceived ease of use.

Participation in Research

- Your participation in this research is entirely voluntary. You have the right to withdraw at any time without giving a reason.
- Withdrawal from the research will not affect any treatment and/or services that you receive.
- You can request your data to be withdrawn and personal information destroyed, as long as it is practical.

Figure 5 - Participant Information Sheet

CONSENT FORM

Title of Study: Usability testing and Evaluation on Google Flights

Lead researcher: Thalal Izzath Anika Mohamed Linushan Sivakumar Saad Mazhar

I have been given the Participation Information Sheet and/or had its contents explained to me.	☐	☐
I have had an opportunity to ask any questions and I am satisfied with the answers given.	☐	☐
I understand I have a right to withdraw from the research at any time and I do not have to provide a reason.	☐	☐
I understand that if I withdraw from the research any data included in the results will be removed if that is practicable (I understand that once anonymised data has been collated into other datasets it may not be possible to remove that data).	☐	☐
I would like to receive information relating to the results from this study.	☐	☐
I wish to receive a copy of this Consent form.	☐	☐
I confirm I am willing to be a participant in the above research study.	☐	☐
I note the data collected may be retained in an archive and I am happy for my data to be reused as part of future research activities. I note my data will be fully anonymised (if applicable).	☐	☐

Participant's Name: _____

Signature: _____ **Date:** _____

This consent form will be stored separately from any data you provide so that your responses remain anonymous.

_____

I confirm I have provided a copy of the Participant Information Sheet approved by the Research Ethics Committee to the participant and fully explained its contents. I have given the participant an opportunity to ask questions, which have been answered.

Researcher's Name: _____

Signature: _____ **Date:** _____

Figure 6 - Participant Information Sheet

Appendix II. Moderator Information Sheet

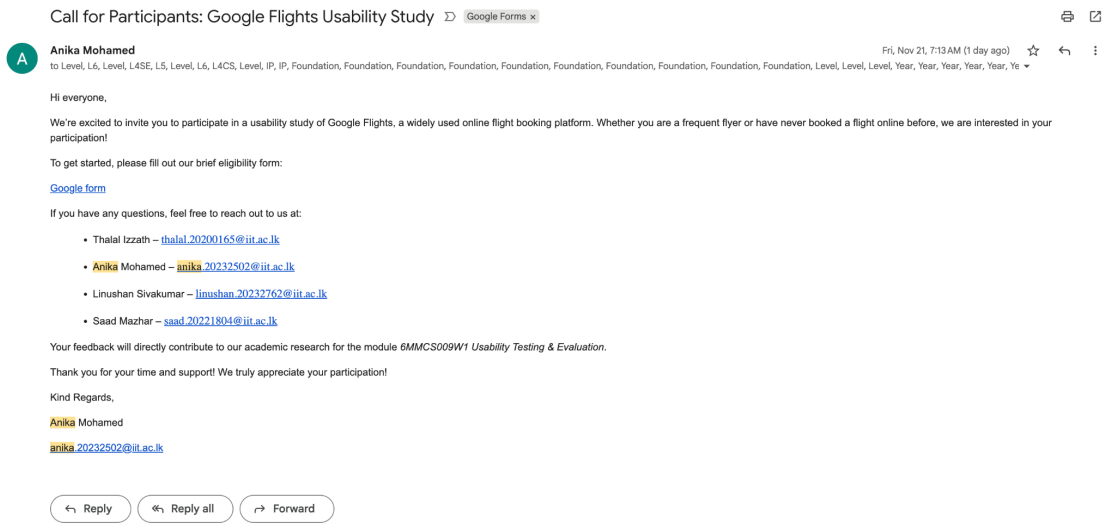
Efficiency (Thalal) - 2.1 - Task Insturctions Sheet (Thalal Izzath)

Effectiveness (Anika) - Instruction Sheet: Effectiveness (Anika)

Error Rate (Saad) - Error-Rate Study - Instruction Sheet

Learnability (Linushan) - 2.4 Learning Intervention.pdf

Appendix III. Participant Recruitment Evidence



Call for Participants: Google Flights Usability Study Google Forms ×

Anika Mohamed Fri, Nov 21, 7:13 AM (1 day ago)

to Level, L6, Level, L4SE, L5, Level, L6, L4CS, Level, IP, IP, Foundation, Foundation, Foundation, Foundation, Foundation, Foundation, Foundation, Foundation, Foundation, Foundation, Level, Level, Level, Year, Year, Year, Year, Year, Ye

Hi everyone,

We're excited to invite you to participate in a usability study of Google Flights, a widely used online flight booking platform. Whether you are a frequent flyer or have never booked a flight online before, we are interested in your participation!

To get started, please fill out our brief eligibility form:

Google form

If you have any questions, feel free to reach out to us at:

- Thalal Izzath – thalal.20200165@iit.ac.lk
- Anika Mohamed – anika.20232502@iit.ac.lk
- Linushan Sivakumar – linushan.20232762@iit.ac.lk
- Saad Mazhar – saad.20221804@iit.ac.lk

Your feedback will directly contribute to our academic research for the module *6MMCS009W1 Usability Testing & Evaluation*.

Thank you for your time and support! We truly appreciate your participation!

Kind Regards,

Anika Mohamed

anika.20232502@iit.ac.lk

Reply Reply all Forward

Figure 7 - Participant Recruitment Evidence

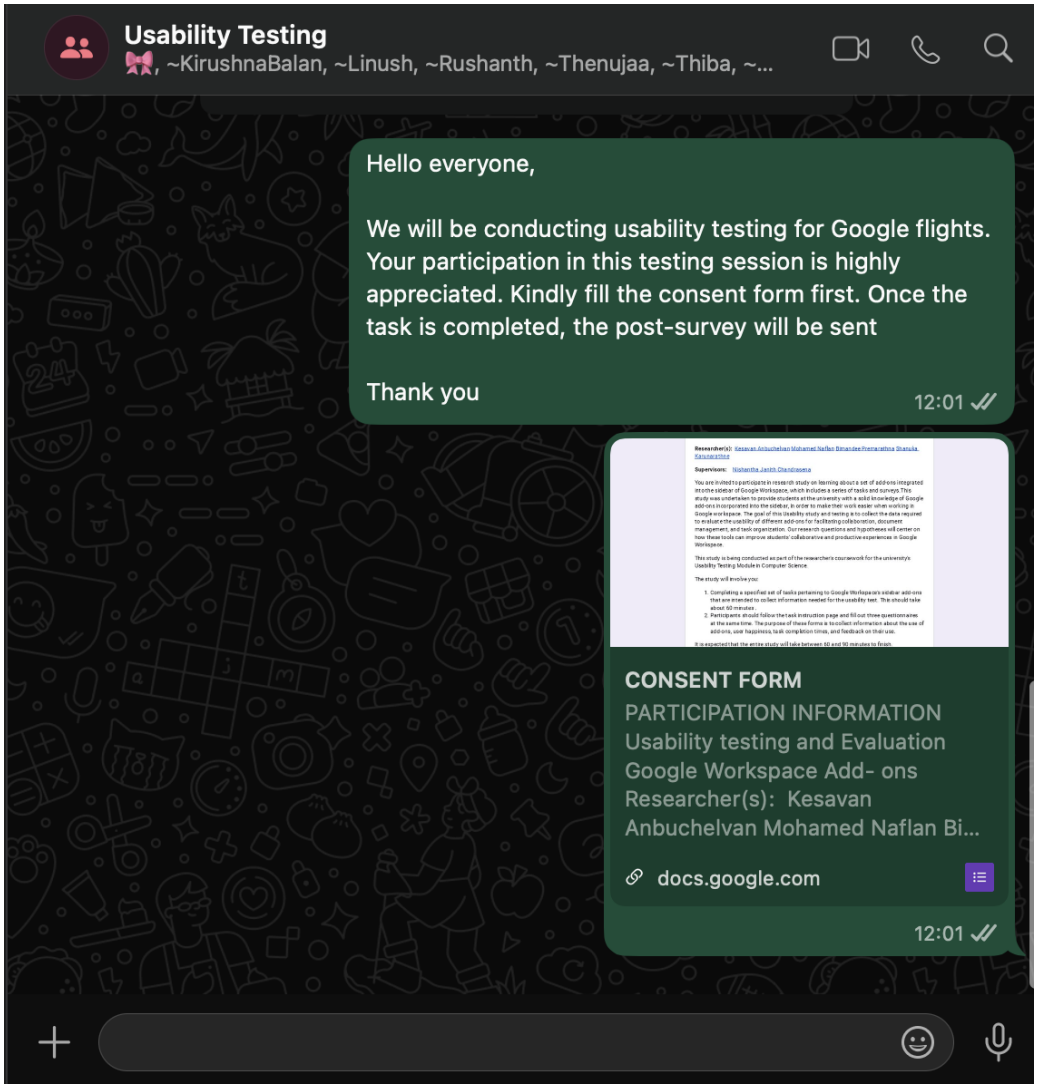


Figure 8 - Participant Recruitment Evidence

Appendix IV. Data Collection Material

https://drive.google.com/drive/folders/1mYP0ngPNCA6s5KALZUC0kLwXb4sF322w?usp=drive_link

IV.A Pre-Test Questionnaire: https://forms.gle/xmfaZmbgk3BrbNmf9

IV.B Post-Test Questionnaire:

1. Efficiency (Thalal) - https://forms.gle/xrcz7DCi9tG67nwGA
2. Effectiveness (Anika) - https://forms.gle/o3ZauShYkxT57tvs8

3. Error Rate (Saad) - https://forms.gle/km58PmUiedqZPVdH7
4. Learnability (Linushan) - https://forms.gle/KswmBuQCAafDw8Vd9

Error Rate checklist - Error Checklist

Error Rate Notes - Error Checklist

IV.C Consent Form: https://forms.gle/uP95BhBQSsCDywa97

IV.A **Pre-Test Questionnaire Form**



Figure 9 - Pre-Test Questionnaire Form

Consent to Participate

Your privacy is of utmost importance to us. We will not be requiring you to disclose your personal information on this questionnaire. If you have disclosed any personal information during this usability test and would like it removed, we assure you that we will promptly delete it upon your request without any inquiries.

Please access the consent form:

If you have any questions or concerns, feel free to contact us:

- *Thalal Izzath - thalal.20200165@iit.ac.lk*
- *Anika Mohamed - anika.20232502@iit.ac.lk*
- *Linushan Sivakumar - linushan.20232762@iit.ac.lk*
- *Saad Mazhar - saad.20221804@iit.ac.lk*

By clicking **Next**, you confirm that you have carefully read and comprehended the terms and are proceeding with your participation under these specified conditions.

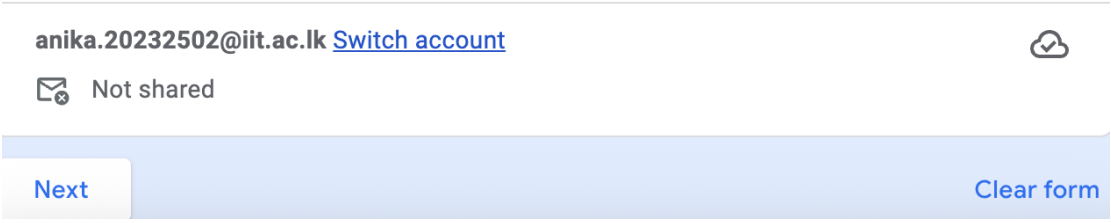


Figure 10 - Pre-Test Questionnaire Form

Pre-Test Questions

- This questionnaire collects important background information to support the analysis of your responses in the usability study. Your honest answers will help us better understand your experience level and ensure the accuracy of our research findings.
- All information you provide will remain strictly confidential and will be used solely for academic and research purposes within this study. There are no right or wrong answers, please respond based on your genuine experience and opinions.
- Your participation in completing all questions is highly appreciated, as it contributes directly to the reliability and validity of our usability evaluation.
- Please note that a laptop or desktop computer is required for the tasks in this test, as Google Flights functions differently on mobile devices.

  Thank you for contributing to this study. Your input is essential and greatly valued.

Full Name *

Your answer

Figure 11 - Pre-Test Questionnaire Form

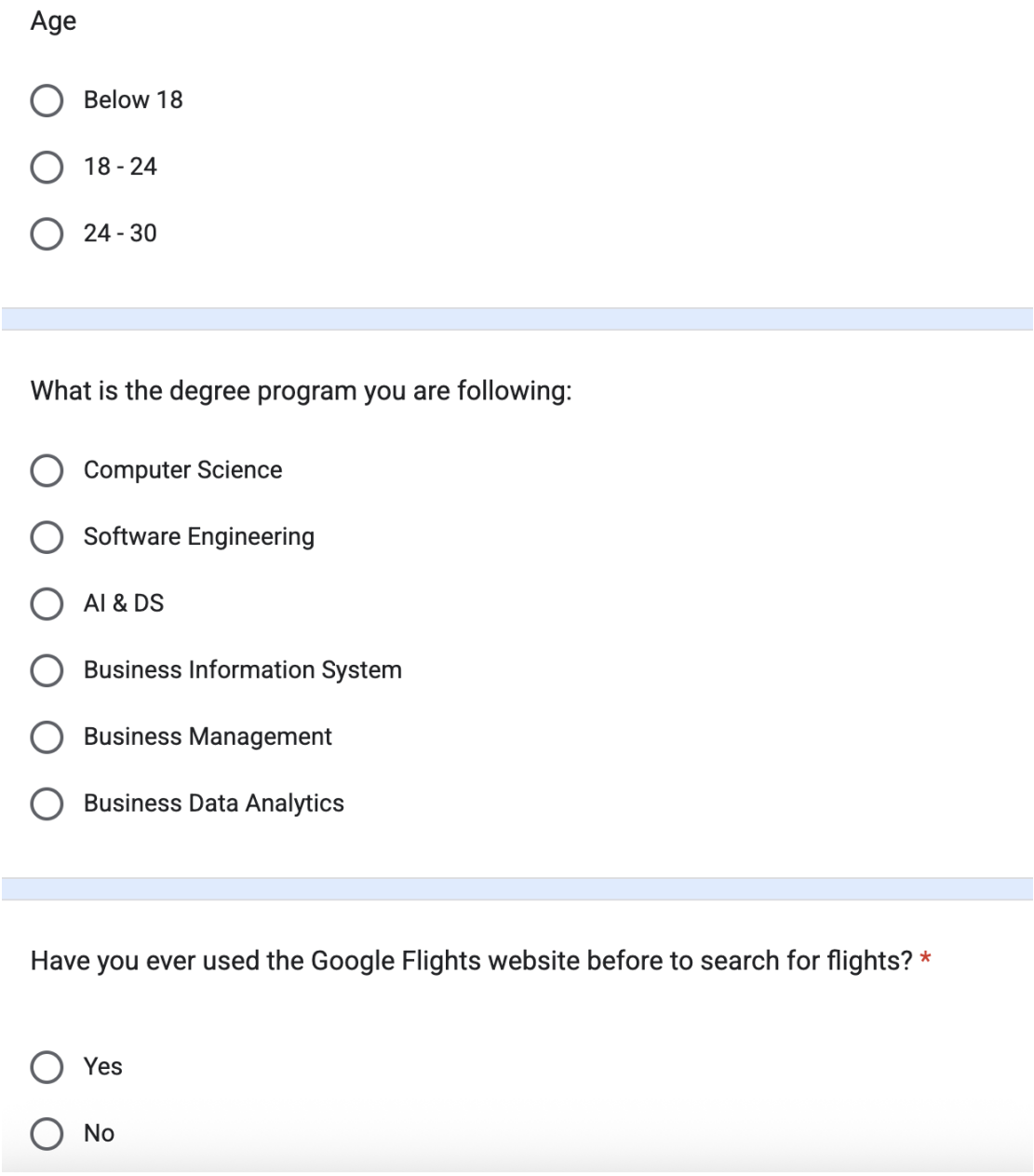


Figure 12 - Pre-Test Questionnaire Form

Which flight-booking platforms have you used before? (Select all that apply) *

- [ ] Google Flights
- [ ] Skyscanner
- [ ] Kayak
- [ ] Airline websites
- [ ] None of the above

Have you used search filters on flight-booking websites before (e.g., Stops, Price, Airlines, Times)? *

- ◯ Yes
- ◯ No
- ◯ Maybe

How confident do you feel in finding the cheapest or best flight deal online? *

	1	2	3	4	5	
Not confident	◯	◯	◯	◯	◯	Very confident

Figure 13 - Pre-Test Questionnaire Form

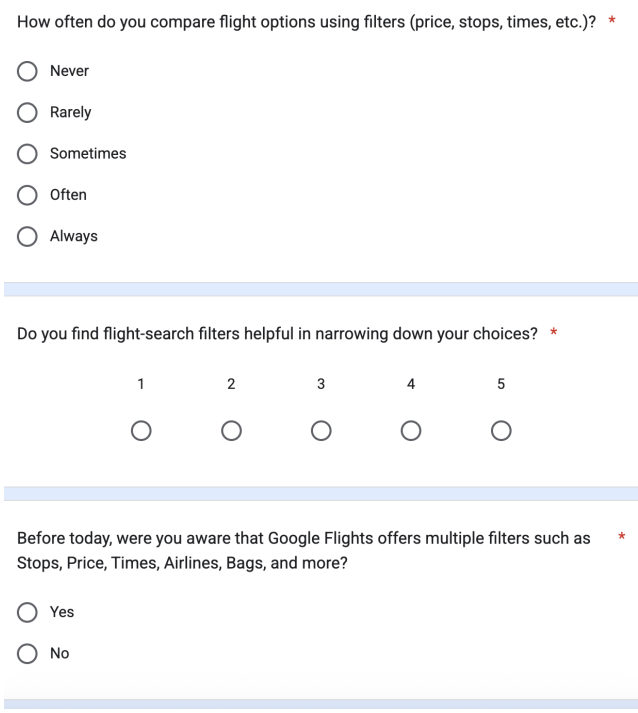


Figure 13 - Pre-Test Questionnaire Form

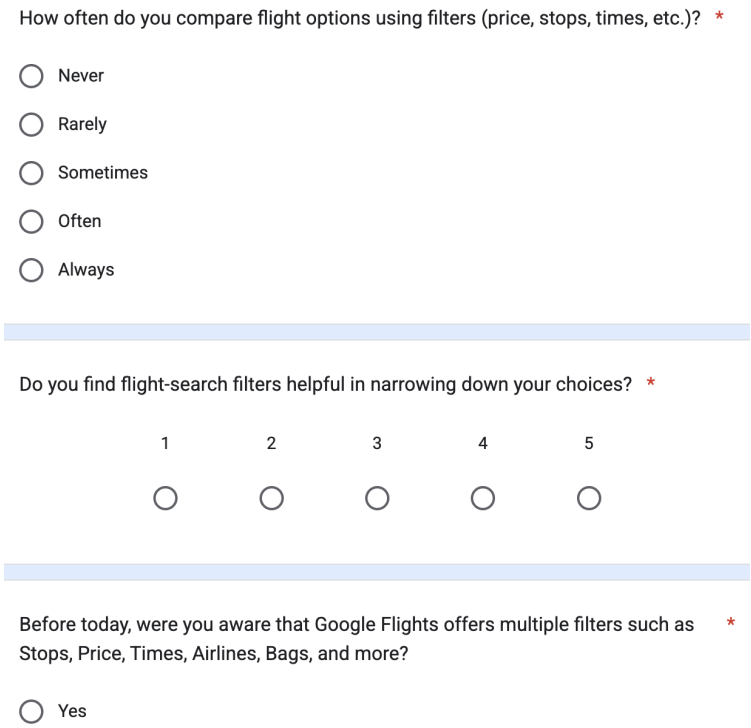
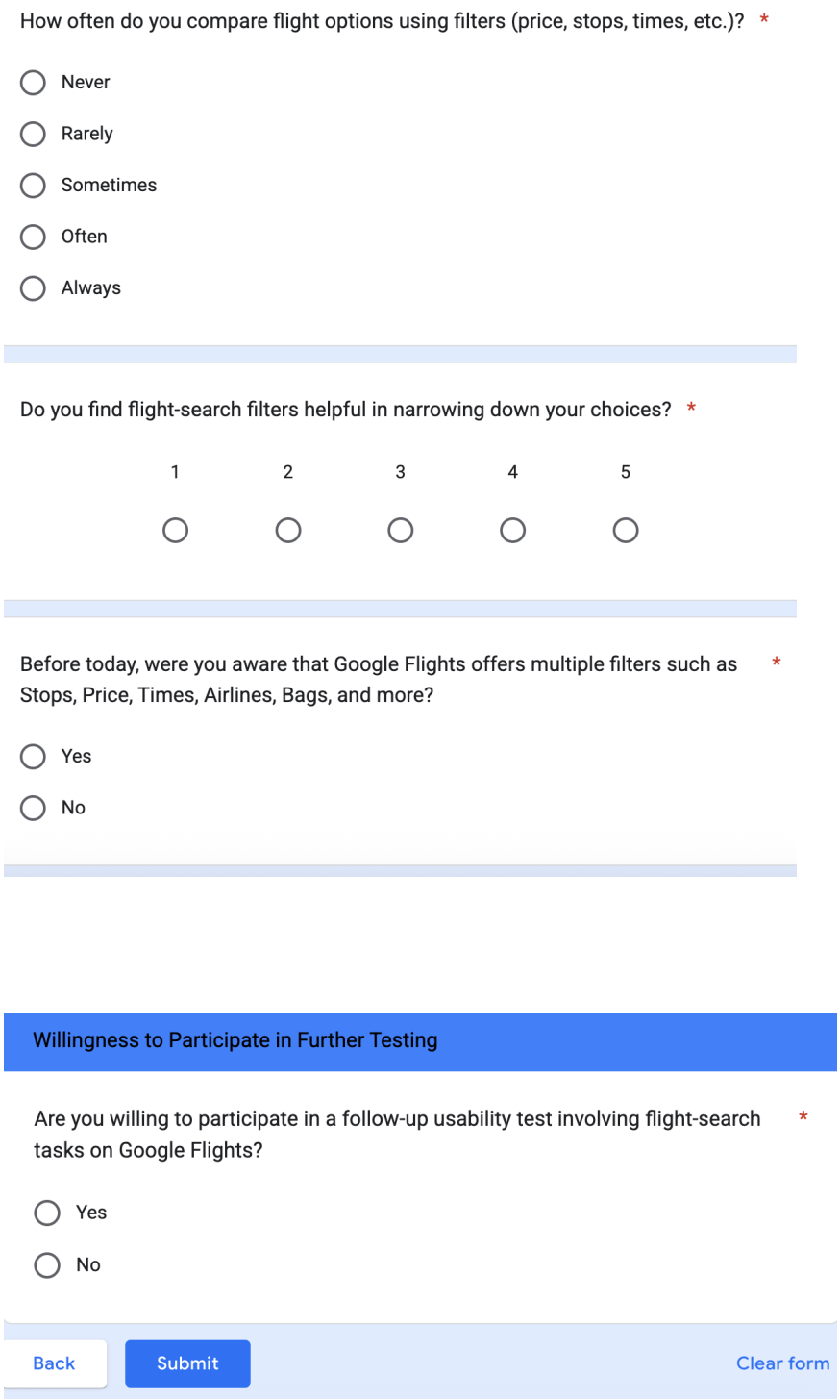


Figure 14 - Pre-Test Questionnaire Form


How often do you compare flight options using filters (price, stops, times, etc.)? *

- Never
- Rarely
- Sometimes
- Often
- Always

Do you find flight-search filters helpful in narrowing down your choices? *

1 2 3 4 5

Before today, were you aware that Google Flights offers multiple filters such as Stops, Price, Times, Airlines, Bags, and more? *

- Yes
- No

Willingness to Participate in Further Testing

Are you willing to participate in a follow-up usability test involving flight-search tasks on Google Flights? *

- Yes
- No

Back Submit Clear form

Figure 15 - Pre-Test Questionnaire Form

CONSENT FORM

PARTICIPATION INFORMATION - CONSENT FORM

Usability Evaluation of the Google Flights Platform

Researchers: Thalal Izzath, Anika Mohamed, Linushan Sivakumar, Saad Mazhar

Supervisor: Ms. Savinthie Suwandaratna

You are being invited to take part in a research study to evaluate the usability of the Google Flights website. Before you decide whether to participate, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully.

What is the purpose of the study? The aim of this research is to assess the Effectiveness, Efficiency, Error Tolerance, and Learnability of Google Flights. We are investigating how users with different levels of experience (from novice to expert) navigate the interface, utilize advanced filtering tools, and complete complex flight booking tasks. The data collected will help identify usability problems and formulate recommendations for improving the user experience.

This study is being conducted as part of the researcher's coursework for the university's Usability Testing Module in Computer Science.

1. Complete a Pre-Test Questionnaire: This collects basic demographic data and information about your prior experience with Google Flights (to categorize your experience level).

Figure 16 - Consent Form

2. Perform Usability Tasks: You will be given specific scenarios. You will perform these tasks on your laptop using the Google Flights website.

3. Allow Recording: Your screen activity and audio will be recorded using screen capture software (e.g., OBS Studio) to allow us to measure time and analyze navigation paths.

4. Complete a Post-Test Questionnaire: After the tasks, you will rate your satisfaction, confidence, and perceived ease of use.

Data Protection and Confidentiality

- Your responses will be anonymized. Your name will not appear in the final report or any public presentation.
- Screen recordings and survey data will be stored on secure, password-protected devices accessible only to the research group and module tutors.
- Data will be retained only for the duration of the academic assessment period and will be destroyed after the Exam Board confirms the results.

Your Rights

- Your participation is entirely voluntary.
- You have the right to withdraw at any time during the session without giving a reason.
- You can refuse to answer any specific question in the surveys.
- If you withdraw, any data collected up to that point will be deleted immediately.

Contact Details If you have any questions about this study, please contact the research team via the email addresses provided by your recruiter, or contact the supervisor via an email.

anika.20232502@iit.ac.lk Switch account

Not shared

* Indicates required question

Participant Name *

Your answer

Date *

Date

yyyy-mm-dd

I confirm that I have read and understood the Participant Information Sheet for the above study. *

Yes

No

Figure 17 - Consent Form

I understand that data collected during this study will be anonymized and used strictly for academic assessment purposes. *

- ◯ Yes
- ◯ No

I understand that I can request for my data to be withdrawn/deleted as long as this is practical (before final submission). *

- ◯ Yes
- ◯ No

I confirm I am willing to be a participant in the above research study. *

- ◯ Yes
- ◯ No

I have had the opportunity to consider the information and ask questions, and I am satisfied with the answers given. *

- ◯ Yes
- ◯ No

I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason. *

- ◯ Yes
- ◯ No

I agree to having my screen activity and audio (voice) recorded whilst performing the tasks for analysis purposes. *

- ◯ Yes
- ◯ No

Figure 18 - Consent Form

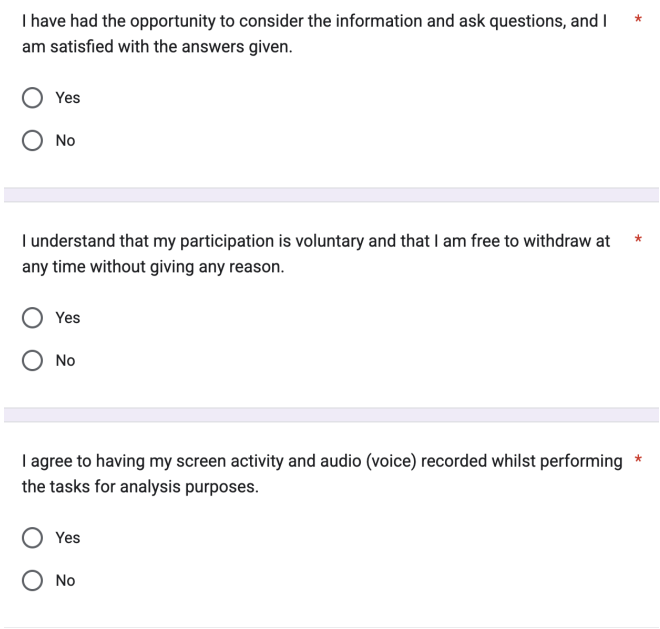


Figure 19 - Consent Form

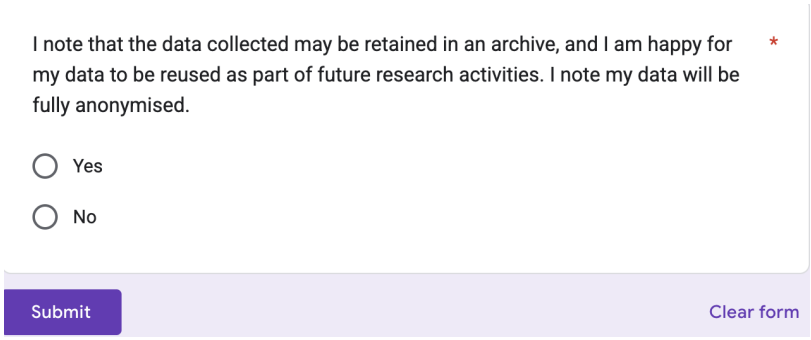


Figure 20 - Consent Form

Appendix IV.B **Post-Test Questionnaire Form**

Post-Test Questionnaire Form Effectiveness - Anika

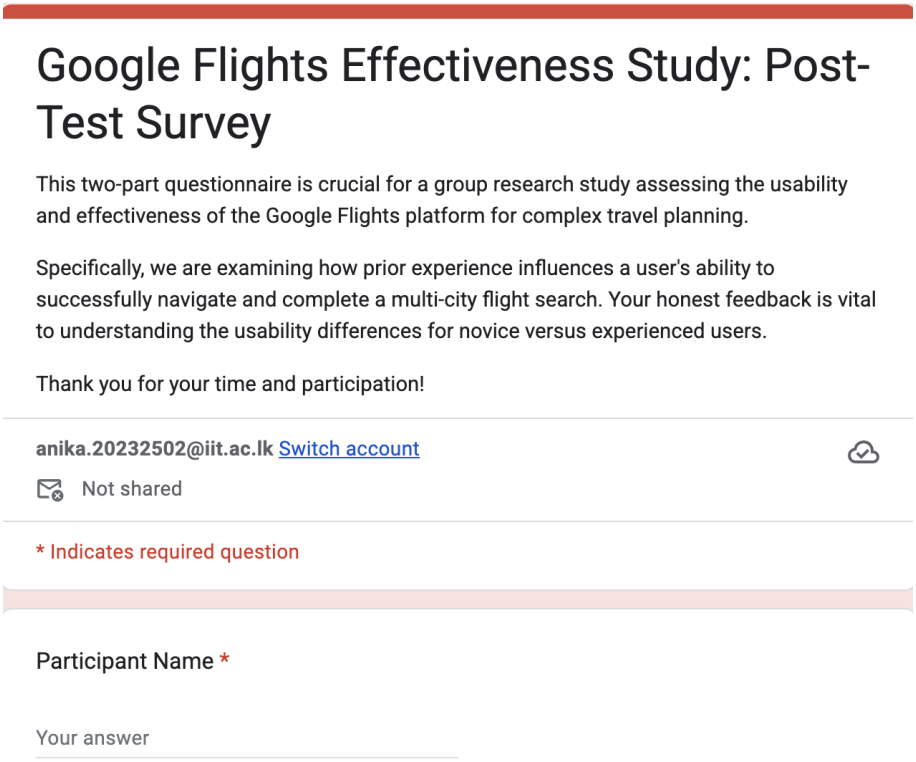


Figure 21 – Post Study Survey Form

Did you successfully find the lowest-cost itinerary matching all the requirements * (3 legs, correct dates, correct cities)?

- ○ Success
- ○ Failure
- ○ I gave up

Overall, how easy was it to find the multi-city search option on Google Flights? *

	1	2	3	4	5	
Very Difficult	○	○	○	○	○	Very Easy

If you failed (answered "Failure" or "I gave up" in Q1), briefly explain the main reason why you could not complete the multi-city search.

Your answer

Figure 22 – Post Study Survey Form

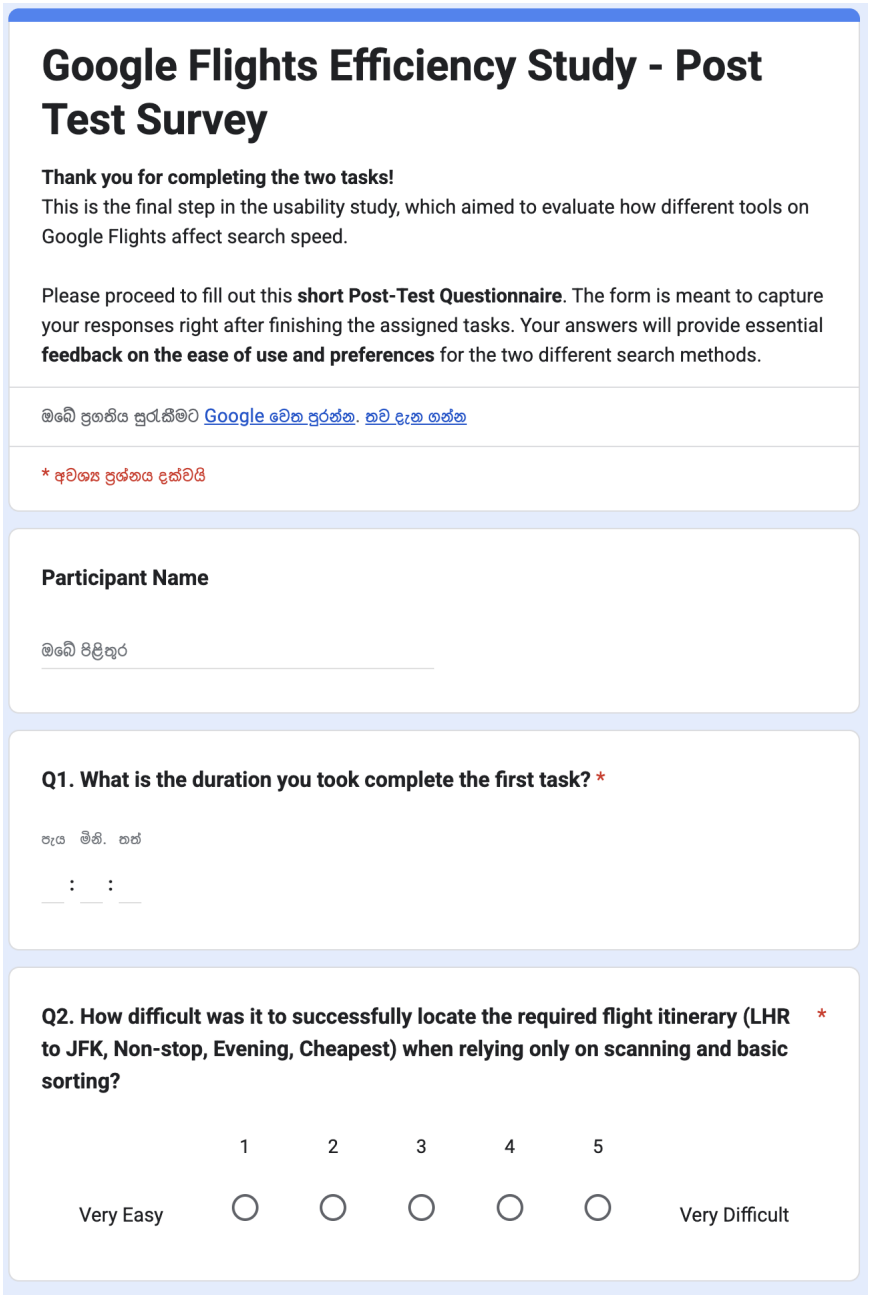
Did you encounter any technical issues (e.g., slow internet, website error) that prevented you from completing the task?

- ○ Yes
- ○ No

Submit Clear form

Figure 23 – Post Study Survey Form

Post-Test Questionnaire Form Efficiency - Thalal



Google Flights Efficiency Study - Post Test Survey

Thank you for completing the two tasks!
This is the final step in the usability study, which aimed to evaluate how different tools on Google Flights affect search speed.

Please proceed to fill out this **short Post-Test Questionnaire**. The form is meant to capture your responses right after finishing the assigned tasks. Your answers will provide essential **feedback on the ease of use and preferences** for the two different search methods.

ඔබේ ප්‍රගතිය සුරැකීමට Google වෙත පුරන්න. තව දැන ගන්න

* අවශ්‍ය ප්‍රශ්නය දක්වයි

Participant Name

ඔබේ පිළිතුර

Q1. What is the duration you took complete the first task? *

පැය මිනි. තත්

: :

Q2. How difficult was it to successfully locate the required flight itinerary (LHR to JFK, Non-stop, Evening, Cheapest) when relying only on scanning and basic sorting? *

	1	2	3	4	5	
Very Easy	○	○	○	○	○	Very Difficult

Figure 24 – Post Study Survey Form

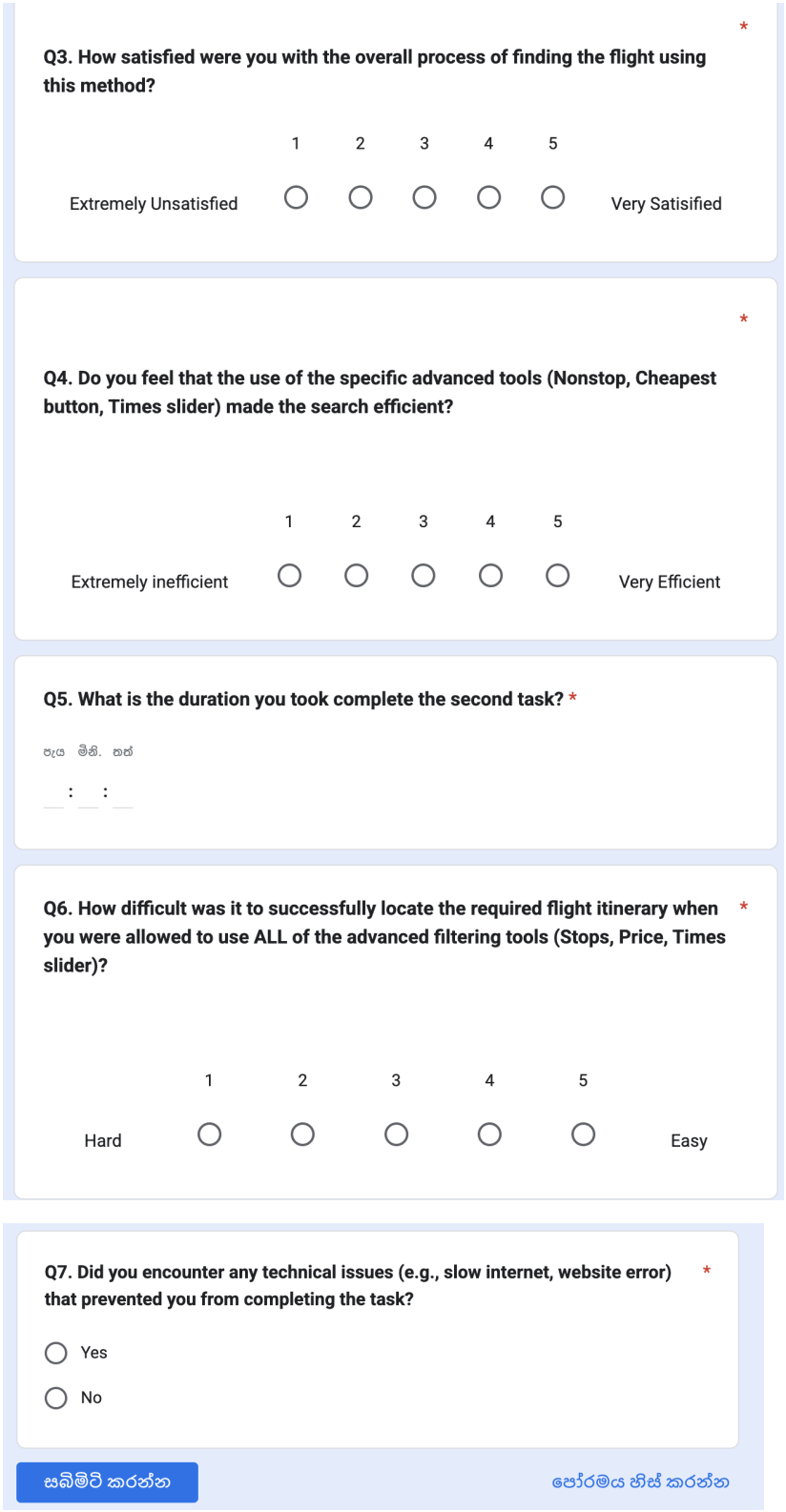
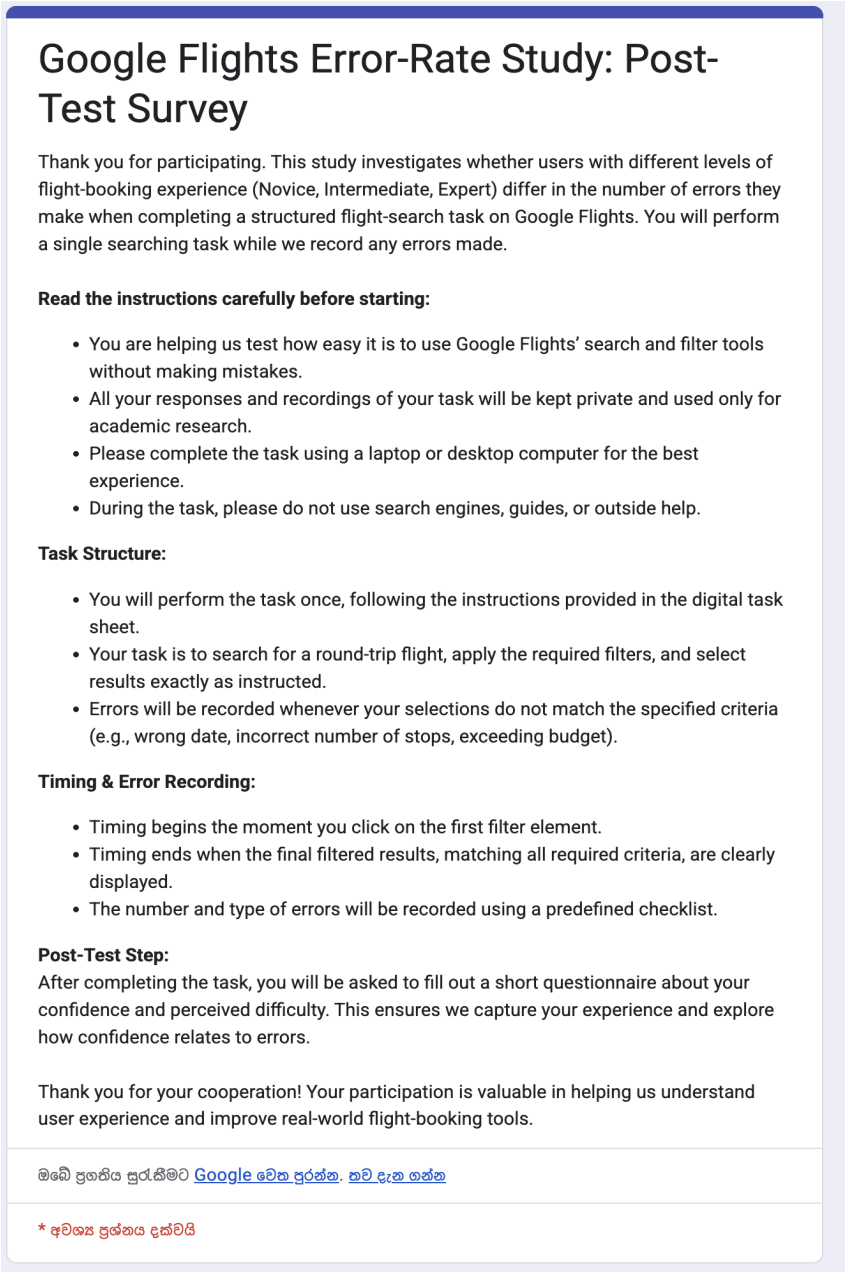


Figure 25 – Post Study Survey Form

Post-Test Questionnaire Form Error Rate - Saad



Google Flights Error-Rate Study: Post-Test Survey

Thank you for participating. This study investigates whether users with different levels of flight-booking experience (Novice, Intermediate, Expert) differ in the number of errors they make when completing a structured flight-search task on Google Flights. You will perform a single searching task while we record any errors made.

Read the instructions carefully before starting:

- You are helping us test how easy it is to use Google Flights' search and filter tools without making mistakes.
- All your responses and recordings of your task will be kept private and used only for academic research.
- Please complete the task using a laptop or desktop computer for the best experience.
- During the task, please do not use search engines, guides, or outside help.

Task Structure:

- You will perform the task once, following the instructions provided in the digital task sheet.
- Your task is to search for a round-trip flight, apply the required filters, and select results exactly as instructed.
- Errors will be recorded whenever your selections do not match the specified criteria (e.g., wrong date, incorrect number of stops, exceeding budget).

Timing & Error Recording:

- Timing begins the moment you click on the first filter element.
- Timing ends when the final filtered results, matching all required criteria, are clearly displayed.
- The number and type of errors will be recorded using a predefined checklist.

Post-Test Step:
After completing the task, you will be asked to fill out a short questionnaire about your confidence and perceived difficulty. This ensures we capture your experience and explore how confidence relates to errors.

Thank you for your cooperation! Your participation is valuable in helping us understand user experience and improve real-world flight-booking tools.

ඔබේ ප්‍රගතිය සුරැකීමට Google වෙත පුරන්න. තව දැන ගන්න

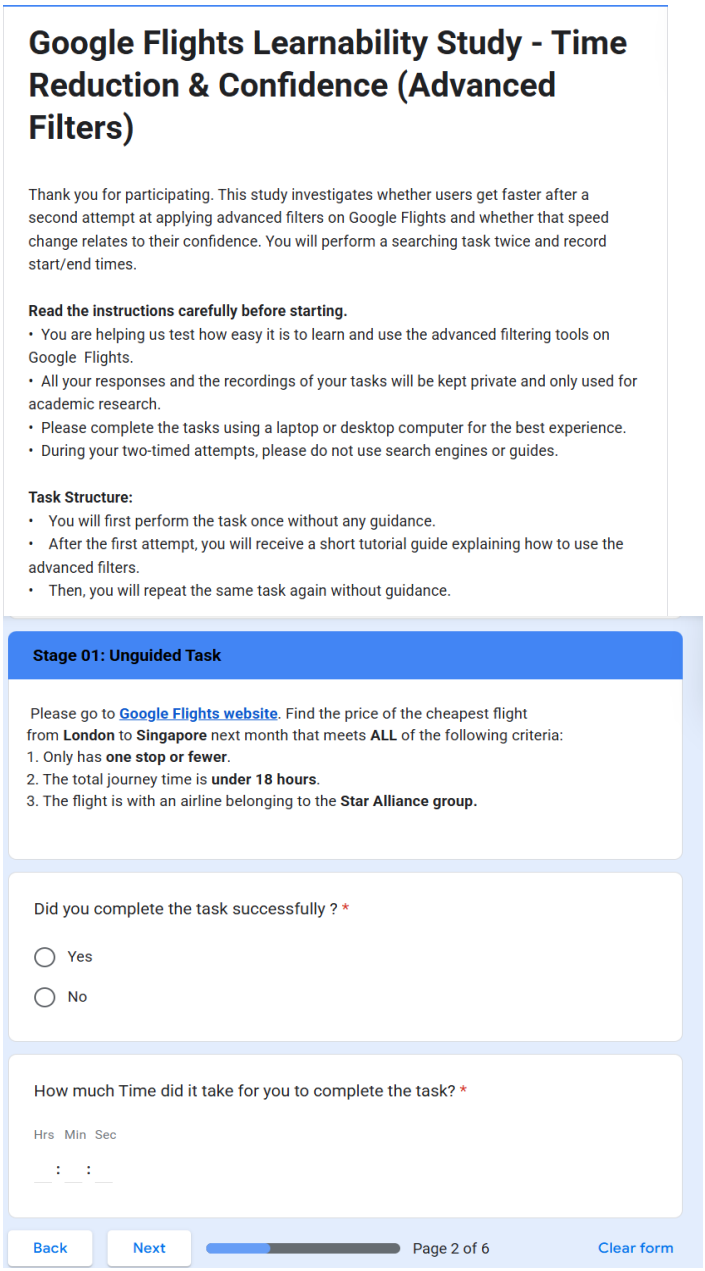
* අවශ්‍ය ප්‍රශ්නය දක්වයි

Figure 26 – Post Study Survey Form



Figure 27 – Post Study Survey Form

Post-Test Questionnaire Form - Learnability – Linushan



Google Flights Learnability Study - Time Reduction & Confidence (Advanced Filters)

Thank you for participating. This study investigates whether users get faster after a second attempt at applying advanced filters on Google Flights and whether that speed change relates to their confidence. You will perform a searching task twice and record start/end times.

Read the instructions carefully before starting.

- You are helping us test how easy it is to learn and use the advanced filtering tools on Google Flights.
- All your responses and the recordings of your tasks will be kept private and only used for academic research.
- Please complete the tasks using a laptop or desktop computer for the best experience.
- During your two-timed attempts, please do not use search engines or guides.

Task Structure:

- You will first perform the task once without any guidance.
- After the first attempt, you will receive a short tutorial guide explaining how to use the advanced filters.
- Then, you will repeat the same task again without guidance.

Stage 01: Unguided Task

Please go to Google Flights website. Find the price of the cheapest flight from **London** to **Singapore** next month that meets **ALL** of the following criteria:
1. Only has **one stop or fewer**.
2. The total journey time is **under 18 hours**.
3. The flight is with an airline belonging to the **Star Alliance group.**

Did you complete the task successfully ? *

- ◯ Yes
- ◯ No

How much Time did it take for you to complete the task? *

Hrs Min Sec

: :

Back Next Page 2 of 6 Clear form

Figure 28 – Post Study Survey Form

Timing Start & End:

- Timing begins the moment you click on the first filter element.
- Timing stops when the final filtered results, matching all required criteria, are clearly displayed.

Post-Test Step:

- After completing the second attempt, you'll be asked to fill out a confident questionnaire. This ensures your natural performance is accurately captured while also measuring how your confidence changes after learning.

Thank you for your cooperation!

* Indicates required question

Your Email *

Your answer

Your Name *

Your answer

Stage 02 : Learning Intervention Guidance

Let me explain how to complete the task.

Learn it !

Guidance

Overview

This tutorial is designed to quickly boost your proficiency in using the advanced filtering tools on Google Flights. Your goal is to master the precise location and application of the three criteria required to efficiently complete the search task. Reviewing this guide is the key step to reducing your task time in the second attempt and demonstrating how easily the platform can be learned.

Figure 29 – Post Study Survey Form

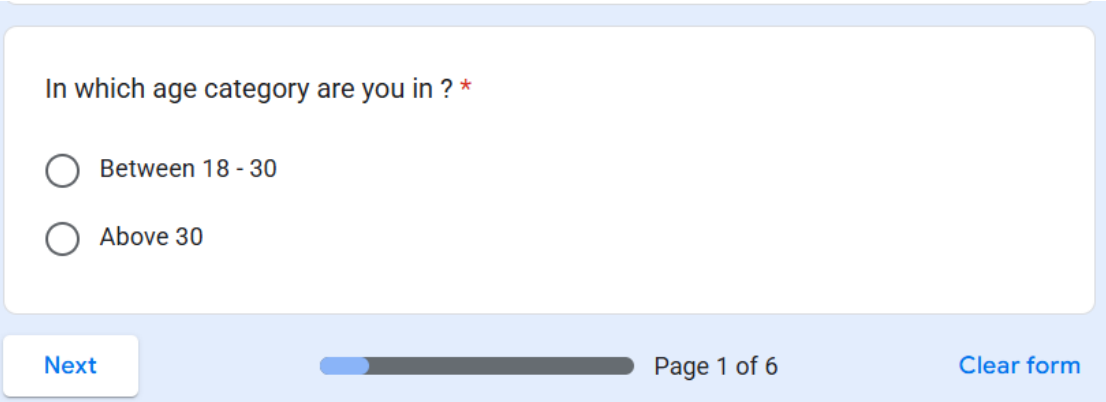


Figure 30 – Post Study Survey Form

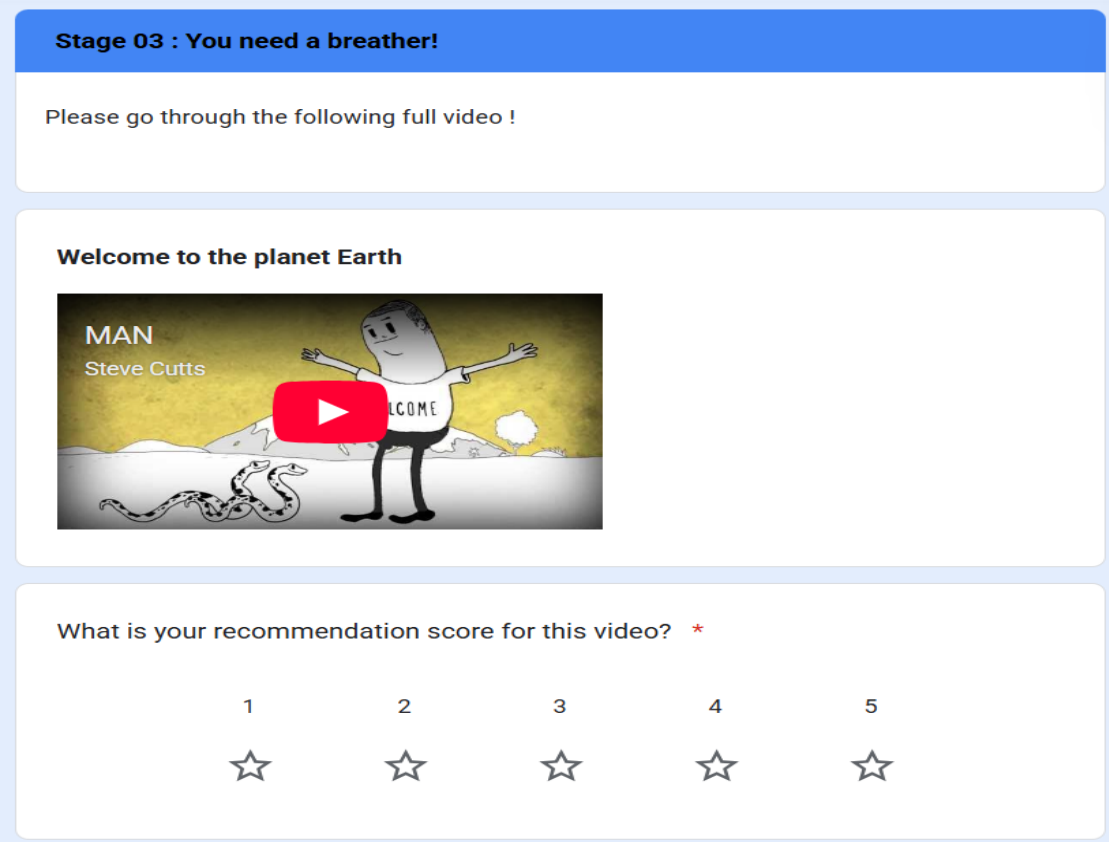


Figure 31 – Post Study Survey Form

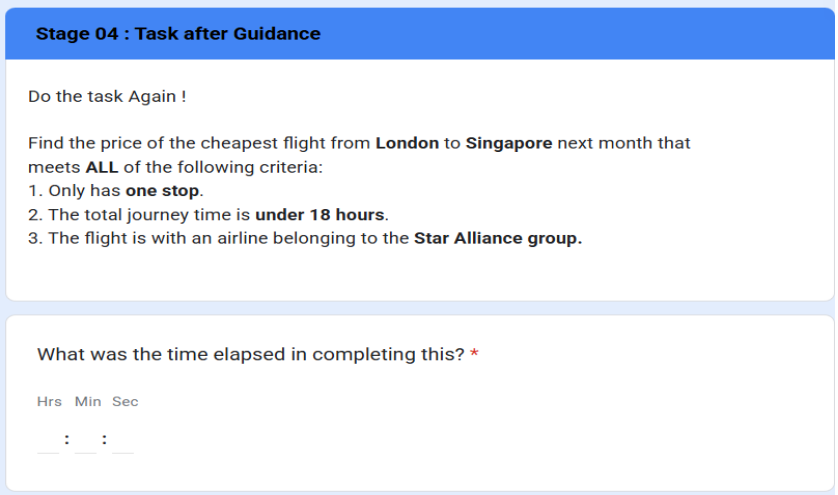


Figure 32 – Post Study Survey Form

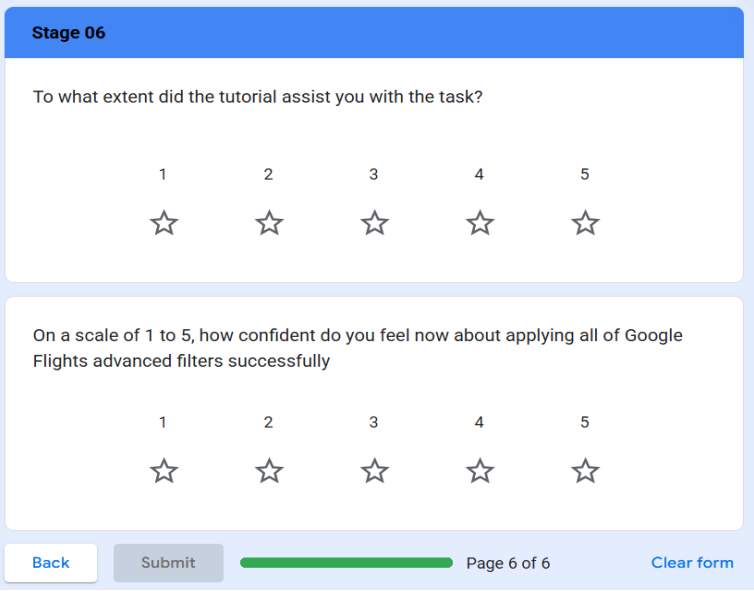


Figure 33 – Post Study Survey Form

Appendix V. Test Recordings

VA. Usability testing session photos



Figure 34 – Post Study Survey Form



Figure 35 – Post Study Survey Form

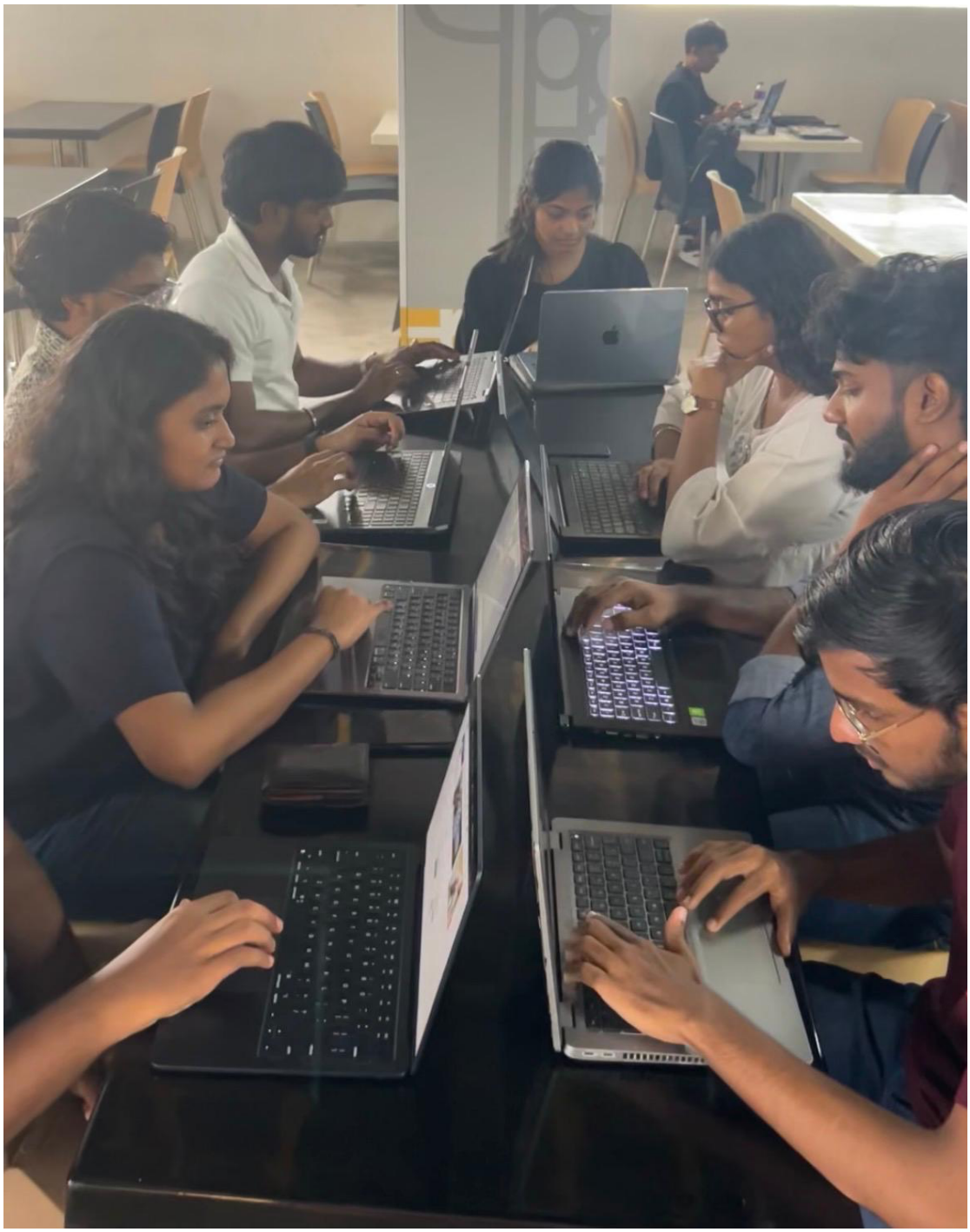


Figure 36 – Post Study Survey Form



Figure 37 – Post Study Survey Form



Figure 38 – Post Study Survey Form

Appendix V.B. Usability testing session Videos Link

https://youtu.be/-XExBBNg4e8

https://youtu.be/_6q_jw8HZc0

https://youtu.be/YA3YzcO9rEY

https://youtube.com/shorts/5p6hgwAqXzI?feature=share

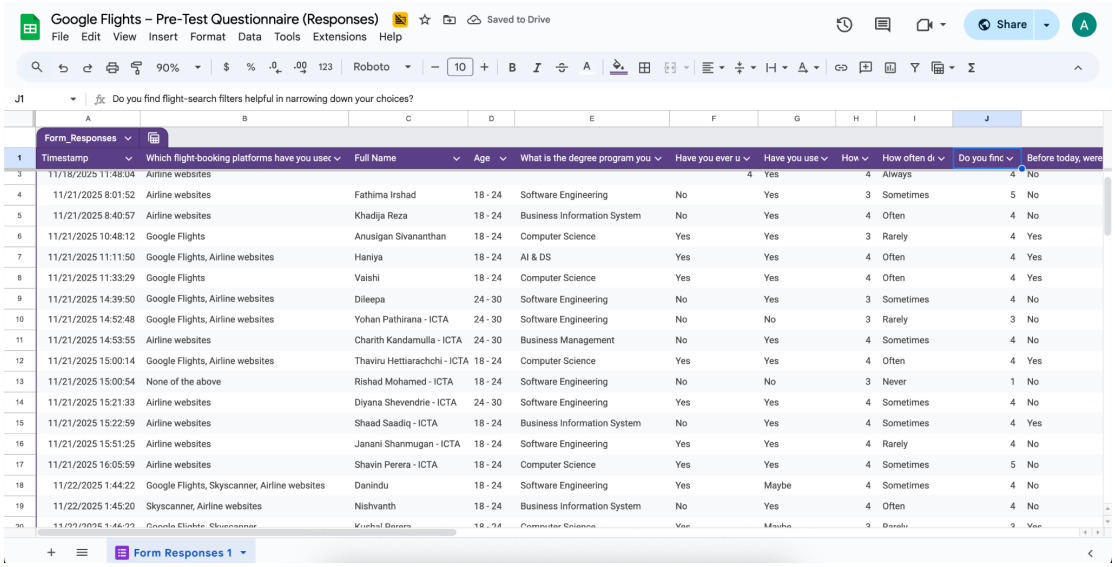
https://youtube.com/shorts/ynx8sy463rI?feature=share

https://drive.google.com/file/d/17Vj7zjKJXKUzOBcs_xwIpzgkAclVHILC/view?usp=drive_link

Recordings - Recordings of questionnaires

Appendix VI. Raw data

Here we have attached links to the Google Form that collects the results from the Google Questionnaire



Timestamp	Which flight-booking platforms have you used	Full Name	Age	What is the degree program you	Have you ever u	Have you use	How	How often d	Do you find	Before today, were
11/18/2025 11:48:04	Airline websites				4	Yes	4	Always	4	No
11/21/2025 8:01:52	Airline websites	Fathima Irshad	18 - 24	Software Engineering	No	Yes	3	Sometimes	5	No
11/21/2025 8:40:57	Airline websites	Khadija Reza	18 - 24	Business Information System	No	Yes	4	Often	4	No
11/21/2025 10:48:12	Google Flights	Anusigan Sivananthan	18 - 24	Computer Science	Yes	Yes	3	Rarely	4	Yes
11/21/2025 11:11:50	Google Flights, Airline websites	Haniya	18 - 24	AI & DS	Yes	Yes	4	Often	4	Yes
11/21/2025 11:33:29	Google Flights	Vaishi	18 - 24	Computer Science	Yes	Yes	4	Often	4	Yes
11/21/2025 14:39:50	Google Flights, Airline websites	Dileepa	24 - 30	Software Engineering	No	Yes	3	Sometimes	4	No
11/21/2025 14:52:48	Google Flights, Airline websites	Yohan Pathirana - ICTA	24 - 30	Software Engineering	No	No	3	Rarely	3	No
11/21/2025 14:53:55	Airline websites	Charith Kandamulla - ICTA	24 - 30	Business Management	No	Yes	4	Sometimes	4	No
11/21/2025 15:00:14	Google Flights, Airline websites	Thaviru Hettiarachchi - ICTA	18 - 24	Computer Science	Yes	Yes	4	Often	4	Yes
11/21/2025 15:00:54	None of the above	Rishad Mohamed - ICTA	18 - 24	Software Engineering	No	No	3	Never	1	No
11/21/2025 15:21:33	Airline websites	Diyana Shevendrie - ICTA	24 - 30	Software Engineering	Yes	Yes	4	Sometimes	4	No
11/21/2025 15:22:59	Airline websites	Shaad Saadiq - ICTA	18 - 24	Business Information System	No	Yes	4	Sometimes	4	Yes
11/21/2025 15:51:25	Airline websites	Janani Shanmugan - ICTA	18 - 24	Software Engineering	Yes	Yes	4	Rarely	4	No
11/21/2025 16:05:59	Airline websites	Shavin Perera - ICTA	18 - 24	Computer Science	Yes	Yes	4	Sometimes	5	No
11/22/2025 1:44:22	Google Flights, Skyscanner, Airline websites	Danindu	18 - 24	Software Engineering	Yes	Maybe	4	Sometimes	4	No
11/22/2025 1:45:20	Skyscanner, Airline websites	Nishvanth	18 - 24	Business Information System	No	Yes	4	Often	4	No

Figure 39 – Pre Test Form Results

Links for the testing results

Pre-Test Questionnaire: Google Flights – Pre-Test Questionnaire (Responses)

Post-Test Questionnaire:

1. Efficiency (Thalal) - 2.2. - Google Flights - Efficiency Survey Form (Responses)
2. Effectiveness (Anika) - Google Flights Usability Study: Post-Test Survey (Responses)
3. Learnability (Linushan) - Learnability - Responses.xlsx
4. Error Rate (Saad)2.3 Google Flights Error-Rate Study: Post-Test Survey (Responses)

Consent Form: CONSENT FORM (Responses)

Usability Testing results Summary: UT Results Summary - Google Flights

Appendix VII. Ethics form

Part A Research-Ethics Consideration Application Form:
Part-A-Research-Ethics-Consideration-Application-form2020_21.doc
Ethics form Cover Sheet: Cover-sheet2020_21(1)